PhenObs Cluster 1 analyses

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Load libraries needed

```
library(readr)
library(here)
```

```
## here() starts at C:/Users/jimenezalfaro/OneDrive - Universidad de Oviedo/IMIB/Analyses/PhenObs_Cluste
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v purrr     1.2.2
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
```

```
## -- Conflicts ---------------------------------------- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(purrr)
library(sf)
```

```
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

```
library(glmmTMB)
```

```
## Warning in check_dep_version(dep_pkg = "TMB"): package version mismatch:
## glmmTMB was built with TMB package version 1.9.21
## Current TMB package version is 1.9.23
## Please re-install glmmTMB from source or restore original 'TMB' package (see '?reinstalling' for more
```

```
library(ggeffects)
library(taxize)
library(rgbif)
library(readxl)
library(gridExtra)
```

```
##
## Adjuntando el paquete: 'gridExtra'
##
## The following object is masked from 'package:dplyr':
##
##     combine
```

```
library(furrr)
```

```
## Cargando paquete requerido: future
##
## Adjuntando el paquete: 'future'
##
## The following object is masked from 'package:taxize':
##
##     resolve
```

```
library(countrycode)
library(progress)
library(ggrepel)
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
library(vegan)
```

```
## Cargando paquete requerido: permute
```

```
library(showtext)
```

```
## Cargando paquete requerido: sysfonts
## Cargando paquete requerido: showtextdb
```

```
library(broom)
library(broom.mixed)
library(reproduciblemanuscript)
```

Set ggplot theme

```
font_add("Times", "C:/Windows/Fonts/times.ttf")
showtext_auto()
```

```
windowsFonts(Times = windowsFont("Times New Roman"))
```

```
theme_set(
  theme_bw(base_family = "Times") +
    theme(
      text = element_text(size = 30),
      axis.title = element_text(size = 30),
      axis.text = element_text(size = 24),
      plot.title = element_text(size = 30)
    )
)
```

Load previously created objects

```
load(here("objects", "sens_per_endanger.Rdata"))
load(here("objects", "sens_per_endanger_stress.Rdata"))
load(here("objects", "sens_per_endanger_stress_ell.Rdata"))
load(here("objects", "sens_threat_Germany.Rdata"))
```

Read data

Red List

Load Red List data (MOTIVATE subset).

```
redlist <- read_tsv(here("data", "raw", "redlist_MOTIVATE_subset.csv"))
```

```
## Rows: 56038 Columns: 26
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
```

```
## chr (23): sPlot concept unified, GBIF concept unified, match_name, original_...
## dbl  (3): V1, accepted_plant_name_id, Publ_year
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
redlist
```

```
## # A tibble: 56,038 x 26
##    `sPlot concept unified` `GBIF concept unified` match_name                V1
##    <chr>                   <chr>                  <chr>                  <dbl>
##  1 Abelia                  Abelia                 Abelia                    NA
##  2 Abelmoschus esculentus  Abelmoschus esculentus Abelmoschus esculentus    NA
##  3 Abies                   Abies                  Abies                     NA
##  4 Abies alba              Abies alba             Abies alba               686
##  5 Abies alba              Abies alba             Abies alba               687
##  6 Abies alba              Abies alba             Abies alba              5088
##  7 Abies alba              Abies alba             Abies alba              8698
##  8 Abies alba              Abies alba             Abies alba             11749
##  9 Abies alba              Abies alba             Abies alba             19497
## 10 Abies alba              Abies alba             Abies alba             23584
## # i 56,028 more rows
## # i 22 more variables: original_species_name <chr>, parsed_species_name <chr>,
## #   accepted_taxon_name <chr>, accepted_taxon_authors <chr>,
## #   original_RL_cat <chr>, Redlist_cat <chr>, Threat <chr>, Country <chr>,
## #   family <chr>, genus <chr>, taxon_rank <chr>, lifeform_description <chr>,
## #   match_type <chr>, match_status <chr>, accepted_plant_name_id <dbl>,
## #   ipni_id <chr>, Publ_year <dbl>, region_name <chr>, ...
```

Sensitivity

Read sensitivity data (slopes from Robert).

```
slopes_FlOn <- read_csv(here("data", "raw", "slopes_20260302", "Overview_Slopes_all_New_FlOn.csv")) %>%
  select(-...1) %>% mutate(stage = "FlOn")
```

```
## New names:
## Rows: 171 Columns: 11
## -- Column specification
## -------------------------------------------------------- Delimiter: "," chr
## (1): Species dbl (10): ...1, mean_slope, ci_lower, ci_upper, slope_var,
## n_gardens, temp_m...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```
slopes_SensOn <- read_csv(here("data", "raw", "slopes_20260302", "Overview_Slopes_all_SenOn.csv")) %>%
  select(-...1) %>% mutate(stage = "SensOn")
```

```
## New names:
## Rows: 156 Columns: 11
```

```
## -- Column specification
## ------------------------------------------------------- Delimiter: "," chr
## (1): Species dbl (10): ...1, mean_slope, ci_lower, ci_upper, slope_var,
## n_gardens, temp_m...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```
slopes_LeafUnf <- read_csv(here("data", "raw", "slopes_20260302", "Overview_Slopes_all_LeafUnf.csv")) %>%
  select(-...1) %>% mutate(stage = "LeafUnf")
```

```
## New names:
## Rows: 175 Columns: 11
## -- Column specification
## ------------------------------------------------------- Delimiter: "," chr
## (1): Species dbl (10): ...1, mean_slope, ci_lower, ci_upper, slope_var,
## n_gardens, temp_m...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```
slopes_GSL <- read_csv(here("data", "raw", "slopes_20260302", "Overview_Slopes_all_GSL.csv")) %>%
  select(-...1) %>% mutate(stage = "GSL")
```

```
## New names:
## Rows: 154 Columns: 11
## -- Column specification
## ------------------------------------------------------- Delimiter: "," chr
## (1): Species dbl (10): ...1, mean_slope, ci_lower, ci_upper, slope_var,
## n_gardens, temp_m...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```
slopes_FrOn <- read_csv(here("data", "raw", "slopes_20260302", "Overview_Slopes_all_New_FrOn.csv")) %>%
  select(-...1) %>% mutate(stage = "FrOn")
```

```
## New names:
## Rows: 149 Columns: 11
## -- Column specification
## ------------------------------------------------------- Delimiter: "," chr
## (1): Species dbl (10): ...1, mean_slope, ci_lower, ci_upper, slope_var,
## n_gardens, temp_m...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```
slopes_FlDur <- read_csv(here("data", "raw", "slopes_20260302", "Overview_Slopes_all_New_FlDur.csv")) %>%
  select(-...1) %>% mutate(stage = "FlDur")
```

```
## New names:
```

```
## Rows: 171 Columns: 11
## -- Column specification
## ------------------------------------------------------- Delimiter: "," chr
## (1): Species dbl (10): ...1, mean_slope, ci_lower, ci_upper, slope_var,
## n_gardens, temp_m...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```
slopes_FlPeak <- read_csv(here("data", "raw", "slopes_20260302", "Overview_Slopes_all_New_FlPeak.csv")) %>%
  select(-...1) %>% mutate(stage = "FlPeak")
```

```
## New names:
## Rows: 169 Columns: 11
## -- Column specification
## ------------------------------------------------------- Delimiter: "," chr
## (1): Species dbl (10): ...1, mean_slope, ci_lower, ci_upper, slope_var,
## n_gardens, temp_m...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```
slopes_FlOn
```

```
## # A tibble: 171 x 11
##    Species    mean_slope ci_lower ci_upper slope_var n_gardens temp_min temp_max
##    <chr>           <dbl>    <dbl>    <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
##  1 Achillea ~     -3.48   -10.5     10.1      26.4          13     5.44     14.6
##  2 Aconitum ~      8.37     1.33    16.9      14.7           6     8.66     12.9
##  3 Acorus ca~     -8.58   -22.9      1.76     41.8           6     9.18     12.0
##  4 Actaea sp~      3.94     0.322    6.65      2.69          4     8.56     12.6
##  5 Adonis ve~     -1.82    -8.15     6.30     13.3           6     8.11     11.1
##  6 Aegopodiu~     -2.69    -6.58     0.416     2.70          7     4.29     12.0
##  7 Ajuga pyr~     -0.622   -4.49    10.7      26.8           3     3.87     10.9
##  8 Alchemill~      6.80    -0.342   18.2      23.5           3     8.73     11.5
##  9 Alchemill~     -6.15   -13.0      1.13     12.8           4     7.76     10.9
## 10 Allium sc~     -5.54    -6.79    -3.78      0.526         5     3.49     14.6
## # i 161 more rows
## # i 3 more variables: temp_mean <dbl>, n_data <dbl>, stage <chr>
```

```
slopes_SensOn
```

```
## # A tibble: 156 x 11
##    Species    mean_slope ci_lower ci_upper slope_var n_gardens temp_min temp_max
##    <chr>           <dbl>    <dbl>    <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
##  1 Achillea ~    -13.9     -23.9     -5.39      27.2         11     8.64     16.2
##  2 Aconitum ~      5.09     -6.79    16.7       28.9          5     8.64     12.4
##  3 Acorus ca~     -7.20    -23.7      5.42      51.7          7     8.64     12.2
##  4 Actaea sp~      9.27     -4.06    20.0       40.8          4     9.85     12.8
##  5 Adonis ve~     -0.935   -16.7     11.0       49.4          6     8.64     12.8
##  6 Aegopodiu~      7.07     -7.94    22.0       51.7          8     4.69     12.2
##  7 Ajuga rep~      4.20     -7.58    12.5       32.0          3    10.4      16.2
```

```
##  8 Alchemill~      6.82    -16.6     34.6      176.          4     8.64     11.6
##  9 Allium sc~     -6.24    -19.7     37.4      171.          3     9.58     15.5
## 10 Allium ur~     -4.86    -11.3      2.70      12.5        10     9.58     15.4
## # i 146 more rows
## # i 3 more variables: temp_mean <dbl>, n_data <dbl>, stage <chr>
```

```
slopes_LeafUnf
```

```
## # A tibble: 175 x 11
##    Species    mean_slope ci_lower ci_upper slope_var n_gardens temp_min temp_max
##    <chr>           <dbl>    <dbl>    <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
##  1 Achillea ~      -2.27   -13.2    8.48       33.1         13     4.86     16.2
##  2 Aconitum ~      -8.12   -13.5   -1.05       10.4          5     8.64     12.8
##  3 Acorus ca~      -7.58   -23.9    0.867      52.0          8     4.83     12.2
##  4 Actaea sp~      13.3      4.78  24.3        26.1          4     8.64     12.4
##  5 Adonis ve~      -6.78   -13.1   -0.0753      9.90         6     8.64     12.8
##  6 Aegopodiu~      -7.69   -20.3    2.98       32.6          7     8.64     12.2
##  7 Ajuga pyr~      -6.38   -12.5   38.5       173.           3     4.69     11.9
##  8 Alchemill~     -15.0    -29.6   -4.52       38.7          4     8.64     11.9
##  9 Allium sc~      -9.17   -12.0   -6.02        2.51         4     3.58     15.5
## 10 Allium ur~      -6.49   -10.7    0.779       7.44        11     4.83     15.5
## # i 165 more rows
## # i 3 more variables: temp_mean <dbl>, n_data <dbl>, stage <chr>
```

```
slopes_GSL
```

```
## # A tibble: 154 x 11
##    Species    mean_slope ci_lower ci_upper slope_var n_gardens temp_min temp_max
##    <chr>           <dbl>    <dbl>    <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
##  1 Achillea ~     -12.2   -22.2       5.00      44.8         11     8.64     15.8
##  2 Aconitum ~      11.7     1.35     20.1       23.0          5     8.64     12.4
##  3 Acorus ca~       3.75  -17.4      20.7       85.6          8     4.83     12.2
##  4 Actaea sp~      -7.48  -24.1      10.9       85.3          4     8.64     12.4
##  5 Adonis ve~       4.57  -10.8      15.3       43.6          6     8.64     12.8
##  6 Aegopodiu~      15.5    -4.36     35.0       84.7          8     4.86     12.2
##  7 Alchemill~      21.7     0.516    48.6      149.           4     8.64     11.6
##  8 Allium sc~      -2.14  -19.3      22.3      169.           3     9.58     15.5
##  9 Allium ur~       1.92   -6.74     12.0       21.8         10     8.64     15.5
## 10 Allium vi~       7.15   -8.55     14.3       28.8          7     4.83     12.2
## # i 144 more rows
## # i 3 more variables: temp_mean <dbl>, n_data <dbl>, stage <chr>
```

```
slopes_FrOn
```

```
## # A tibble: 149 x 11
##    Species    mean_slope ci_lower ci_upper slope_var n_gardens temp_min temp_max
##    <chr>           <dbl>    <dbl>    <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
##  1 Achillea ~     -9.82   -31.2      0.898     58.6         11    10.1      17.4
##  2 Aconitum ~      6.12    -0.795   15.0       17.1          6    10.1      13.9
##  3 Acorus ca~     12.4     -4.53    27.2       52.5          7    10.1      13.5
##  4 Actaea sp~      1.70    -4.04     7.51       9.89         4     9.25     13.6
##  5 Adonis ve~     -1.26    -6.76     3.94       7.87         6     8.93     12.2
```

```
##  6 Aegopodiu~     -3.48   -13.6      3.29      18.2          6    10.4      13.1
##  7 Alchemill~      2.08    -4.29     9.52      13.0          3    11.0      12.9
##  8 Alchemill~      0.905   -5.78     5.96       9.14         4    10.2      12.2
##  9 Allium sc~     -2.51    -6.03     2.24       4.15         5     4.95     15.0
## 10 Allium ur~     -5.54    -9.67     1.98       9.94        10     8.93     14.8
## # i 139 more rows
## # i 3 more variables: temp_mean <dbl>, n_data <dbl>, stage <chr>
```

```
slopes_FlDur
```

```
## # A tibble: 171 x 11
##    Species    mean_slope ci_lower ci_upper slope_var n_gardens temp_min temp_max
##    <chr>           <dbl>    <dbl>    <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
##  1 Achillea ~      11.6     -2.62   24.8       46.4         11     7.9      16.7
##  2 Aconitum ~      -3.34   -13.0     3.22      15.2          6     7.28     10.9
##  3 Acorus ca~       5.55    -4.58   14.8       24.5          6     7.89     10.8
##  4 Actaea sp~     -13.2    -18.9    -5.26      10.8          4     8.22     11.9
##  5 Adonis ve~      -6.83   -13.4    -1.31       9.45         6     8.23     11.1
##  6 Aegopodiu~      -2.37    -6.28   -0.593      1.95         7     3.65     10.8
##  7 Agrimonia~      -4.52    -8.18   -1.19       2.98         3     6.83     13.6
##  8 Ajuga pyr~       1.47    -2.35    6.75       7.98         3     4        11.4
##  9 Alchemill~      58.0     43.8    77.5       67.9          4     8.62     12.6
## 10 Allium sc~      15.4     11.3    19.2        4.16         5     2.53     16.5
## # i 161 more rows
## # i 3 more variables: temp_mean <dbl>, n_data <dbl>, stage <chr>
```

```
slopes_FlPeak
```

```
## # A tibble: 169 x 11
##    Species    mean_slope ci_lower ci_upper slope_var n_gardens temp_min temp_max
##    <chr>           <dbl>    <dbl>    <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
##  1 Achillea ~      -3.02   -9.68    8.31      20.1          12     9.74     16.2
##  2 Aconitum ~       7.46    2.72   12.6        7.30          6     9.18     13.6
##  3 Acorus ca~      -8.65  -17.6    -0.551     18.4          6     8.25     11.6
##  4 Actaea sp~       1.02   -4.56    4.21       4.74         4     7.57     12.6
##  5 Adonis ve~      -3.66  -10.3     4.63      14.2          6     8        11.0
##  6 Aegopodiu~      -3.21   -6.13    0.719      3.32         7     5.12     12.0
##  7 Alchemill~       5.93   -0.623  14.2       14.8          3     8.91     11.8
##  8 Alchemill~      -3.10   -8.00    3.90      10.4          4     8.25     10.9
##  9 Allium sc~      -3.87   -5.24   -2.57       0.456        5     4.05     14.9
## 10 Allium ur~      -5.95   -9.50    0.0617     5.69        11     5.31     14.7
## # i 159 more rows
## # i 3 more variables: temp_mean <dbl>, n_data <dbl>, stage <chr>
```

Join and get unique species

```
slopes_all <- bind_rows(slopes_FlOn, slopes_SensOn, slopes_LeafUnf, slopes_GSL,
                        slopes_FrOn, slopes_FlDur, slopes_FlPeak)
```

```
slopes_all_species <- slopes_all %>% distinct(Species)
```

Stress tolerance

Read stress tolerance scores.

```
stress <- read_delim(here("data", "raw", "Stress_Tolerance.csv"), delim = ";") %>%
  rename(stress_tol = `Stress tolerator score_percentage`)
```

```
## Rows: 263 Columns: 2
## -- Column specification --------------------------------------------------------
## Delimiter: ";"
## chr (1): Species
## dbl (1): Stress tolerator score_percentage
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
stress
```

```
## # A tibble: 263 x 2
##    Species               stress_tol
##    <chr>                      <dbl>
##  1 Achillea millefolium        44.4
##  2 Aconitum lycoctonum         13.4
##  3 Acorus calamus              23.8
##  4 Actaea spicata              11.1
##  5 Adonis amurensis            24.4
##  6 Adonis vernalis             41.5
##  7 Aegopodium podagraria        6.7
##  8 Agrimonia eupatoria         41.6
##  9 Agrimonia pilosa            53.7
## 10 Ajuga pyramidalis           34.1
## # i 253 more rows
```

Ellenberg tendency of change

Read Ellenberg tendency of change data.

```
ellenberg <- read_excel(here("data", "raw", "IndicatorValues_Ellenberg.xlsx"))
```

```
## Warning: Expecting numeric in Q1081 / R1081C17: got '?'
```

```
## Warning: Expecting numeric in Q1287 / R1287C17: got '?'
```

```
## Warning: Expecting numeric in Q2746 / R2746C17: got '?'
```

```
ellenberg
```

```
## # A tibble: 2,792 x 19
##    NR_E  Art    GATT_NAME OEK_L OEK_T OEK_K OEK_F OEK_R OEK_N OEK_S LF_LEB LF_B
##    <chr> <chr>  <chr>     <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>  <chr>
##  1 101   Abies~ Abies     (3)   5     4     x     x     x     0     P      I
##  2 401   Acer ~ Acer      (5)   6     4     5     7     6     0     P      S
##  3 403   Acer ~ Acer      (6)   8     4     3     8     4     0     P      S
##  4 404   Acer ~ Acer      (5)   6     6     6     7     7     0     P      S
##  5 405   Acer ~ Acer      (5)   8     4     4     8     6     0     P      S
##  6 408   Acer ~ Acer      (4)   6     4     x     x     x     0     P      S
##  7 409   Acer ~ Acer      (4)   x     4     6     x     7     0     P      S
##  8 501   Acera~ Aceras    7     7     2     4~    8     3     0     G      S
##  9 601   Achil~ Achillea  9     2     4     5     8     3     0     H      W
## 10 623   Achil~ Achillea  8     6     6     8~    7     6     0     H      S
## # i 2,782 more rows
## # i 7 more variables: SZ_VERH_GR <chr>, SZ_VERH_K <chr>, SZ_O <dbl>,
## #   SZ_V <dbl>, HAEUF_M <dbl>, HAEUF_D <chr>, HAEUF_AE_change <dbl>
```

Resolve species names

Resolve species in all datasets to GBIF accepted names using the package taxize:

Species in sensitivity data

Species in Red List data

Resolve species (`sPlot concept unified`) to GBIF accepted names:

Species in stress tolerance scores

Species in Ellenberg tendency of change data

```
ellenberg <- ellenberg %>%
  mutate(Art = ifelse(Art == "Primula veris (officinalis)#f)",
                      "Primula veris (officinalis)", Art))
```

Create occurrence matrix

Occurrence matrices are created using the package rgbif.

Get countries form RL

We want to get an occurrence matrix for all countries in Europe that are included in our Red List data.

```r
# Inspect all countries present in Red List sample
redlist %>% distinct(Country) %>% arrange(Country) %>% print(n = 50)
```

```
## # A tibble: 43 x 1
##    Country
##    <chr>
##  1 Albania
##  2 Andorra
##  3 Austria
##  4 Belarus
##  5 Belgium
##  6 Bosnia and Herzegovina
##  7 Bulgaria
##  8 Croatia
##  9 Cyprus
## 10 Czech Republic
## 11 Denmark
## 12 Estonia
## 13 Finland
## 14 France
## 15 Germany
## 16 Greece
## 17 Hungary
## 18 Ireland
## 19 Italy
## 20 Kosovo
## 21 Latvia
## 22 Liechtenstein
## 23 Lithuania
## 24 Luxembourg
## 25 Malta
## 26 Moldova
## 27 Montenegro
## 28 Netherlands
## 29 Norway
## 30 Poland
## 31 Portugal
## 32 Romania
## 33 Russia
## 34 Serbia
## 35 Slovakia
## 36 Slovenia
## 37 Spain
## 38 Sweden
## 39 Switzerland
## 40 Turkey
## 41 Ukraine
## 42 United Kingdom
## 43 <NA>
```

Get vector of countries (remove NA).

```
countries_raw <- redlist %>%
  distinct(Country) %>%
  filter(!is.na(Country)) %>%
  pull(Country)
```

Convert country names to ISO2 country codes (used by rgbif).

```
# Custom mapping for tricky names
# - GBIF wants 'GB' for United Kingdom / Great Britain
# - 'Czech Republic' -> 'CZ'
# - 'Bosnia and Herzegovina' -> 'BA'
# - 'Russia' -> 'RU'
# - 'Kosovo' -> 'XK' (unofficial but accepted by GBIF)
custom_match <- c(
  "United Kingdom"        = "GB",
  "Great Britain"         = "GB",
  "Czech Republic"        = "CZ",
  "Bosnia and Herzegovina"= "BA",
  "Russia"                = "RU",
  "Kosovo"                = "XK"
)

# Convert to ISO-2 (countrycode handles the rest)
countries_iso <- tibble(
  country_name = countries_raw,
  iso2 = countrycode(countries_raw,
                     origin = "country.name",
                     destination = "iso2c",
                     custom_match = custom_match)
) %>%
  arrange(country_name)
countries_iso
```

```
## # A tibble: 42 x 2
##    country_name           iso2
##    <chr>                  <chr>
##  1 Albania                AL
##  2 Andorra                AD
##  3 Austria                AT
##  4 Belarus                BY
##  5 Belgium                BE
##  6 Bosnia and Herzegovina BA
##  7 Bulgaria               BG
##  8 Croatia                HR
##  9 Cyprus                 CY
## 10 Czech Republic         CZ
## # i 32 more rows
```

```
# Final clean vector for rgbif
gbif_iso2 <- countries_iso %>%
  filter(!is.na(iso2)) %>%
  distinct(iso2) %>%
```

```
  pull(iso2)

gbif_iso2
```

```
##  [1] "AL" "AD" "AT" "BY" "BE" "BA" "BG" "HR" "CY" "CZ" "DK" "EE" "FI" "FR" "DE"
## [16] "GR" "HU" "IE" "IT" "XK" "LV" "LI" "LT" "LU" "MT" "MD" "ME" "NL" "NO" "PL"
## [31] "PT" "RO" "RU" "RS" "SK" "SI" "ES" "SE" "CH" "TR" "UA" "GB"
```

Build occurrence matrix

Build the species × country occurrence matrix with rgbif.

Get species and countries vectors.

Use parallel GBIF querying.

Build species × country combinations.

Checks.

Build the presence/absence matrix.

Matching Phenobs - MOTIVATE Red List data

Do not only use the RL of the countries where we have our botanical gardens, instead use the RL from all available countries in Europe!

Try to match the full PhenObs spp list with the RL to see how many species are there in each category for each country.

Add ISO2 country codes to RL

Join RL and spp list in sensitivity data from PhenObs

Data prep RL

Cases where there is info for more than one year for a species and country:

Keep threat info for the latest publication year. In Russia, there is no info about publication year.

Keep Russia separated:

Remove Russia:

For other countries apart from Russia: If there are several rows for same publication year, pick the most severe red list category.

Check that there is only one entry for each species and country:

For Russia:

For Russia, check that there is only one entry for each species:

Join noRussia and Russia to get final RL for PhenObs:

Join occurrence data and Red List data

Calculate % endangerment

Data prep sensitivity data

Prepare datasets for models

Sensitivity + % endangerment

Sensitivity + % endangerment + stress

Sensitivity + % endangerment + Ellenberg tendency of change

Deal with non-numeric values in indicator values

Columns OEK_L - OEK_S and HAEUF_AE_change

```
ellenberg <- ellenberg %>%
  mutate(across(OEK_L:OEK_S, ~ parse_number(.)))
```

```
## Warning: There were 7 warnings in `mutate()`.
## The first warning was:
## i In argument: `across(OEK_L:OEK_S, ~parse_number(.))`.
## Caused by warning:
## ! 28 parsing failures.
## row col expected actual
## 184  -- a number      x
## 200  -- a number      x
## 676  -- a number      x
## 744  -- a number      x
## 798  -- a number      x
## ... ... ........ ......
## See problems(...) for more details.
## i Run `dplyr::last_dplyr_warnings()` to see the 6 remaining warnings.
```

Join with sens_per_endanger:

Save datasets for later use

Reshape data

```
sens_endanger <- sens_per_endanger %>%
  pivot_longer(cols = matches("^(mean_slope|slope_var|sens_01|sens_sign)_"),
               names_to = c(".value", "stage"),
               names_pattern = "(mean_slope|slope_var|sens_01|sens_sign)_(.*)")
sens_endanger
```

```
## # A tibble: 1,302 x 9
##    species              total_countries threatened_countries perc_endangerment
##    <chr>                          <int>                <int>              <dbl>
##  1 Achillea millefolium              40                    0                  0
##  2 Achillea millefolium              40                    0                  0
##  3 Achillea millefolium              40                    0                  0
##  4 Achillea millefolium              40                    0                  0
##  5 Achillea millefolium              40                    0                  0
##  6 Achillea millefolium              40                    0                  0
##  7 Achillea millefolium              40                    0                  0
##  8 Aconitum lycoctonum               36                    3                  8.33
##  9 Aconitum lycoctonum               36                    3                  8.33
## 10 Aconitum lycoctonum               36                    3                  8.33
## # i 1,292 more rows
## # i 5 more variables: prop_endangerment <dbl>, stage <chr>, mean_slope <dbl>,
## #   slope_var <dbl>, sens_01 <fct>
```

```
sens_endanger_stress <- sens_per_endanger_stress %>%
  pivot_longer(cols = matches("^(mean_slope|slope_var|sens_01|sens_sign)_"),
               names_to = c(".value", "stage"),
               names_pattern = "(mean_slope|slope_var|sens_01|sens_sign)_(.*)")
sens_endanger_stress
```

```
## # A tibble: 1,302 x 10
##    species              total_countries threatened_countries perc_endangerment
##    <chr>                          <int>                <int>              <dbl>
##  1 Achillea millefolium              40                    0                  0
##  2 Achillea millefolium              40                    0                  0
##  3 Achillea millefolium              40                    0                  0
##  4 Achillea millefolium              40                    0                  0
##  5 Achillea millefolium              40                    0                  0
##  6 Achillea millefolium              40                    0                  0
##  7 Achillea millefolium              40                    0                  0
##  8 Aconitum lycoctonum               36                    3                  8.33
##  9 Aconitum lycoctonum               36                    3                  8.33
## 10 Aconitum lycoctonum               36                    3                  8.33
## # i 1,292 more rows
## # i 6 more variables: prop_endangerment <dbl>, stress_tol <dbl>, stage <chr>,
## #   mean_slope <dbl>, slope_var <dbl>, sens_01 <fct>
```

```
sens_endanger_stress_ell <- sens_per_endanger_stress_ell %>%
  pivot_longer(cols = matches("^(mean_slope|slope_var|sens_01|sens_sign)_"),
               names_to = c(".value", "stage"),
               names_pattern = "(mean_slope|slope_var|sens_01|sens_sign)_(.*)")
sens_endanger_stress_ell
```

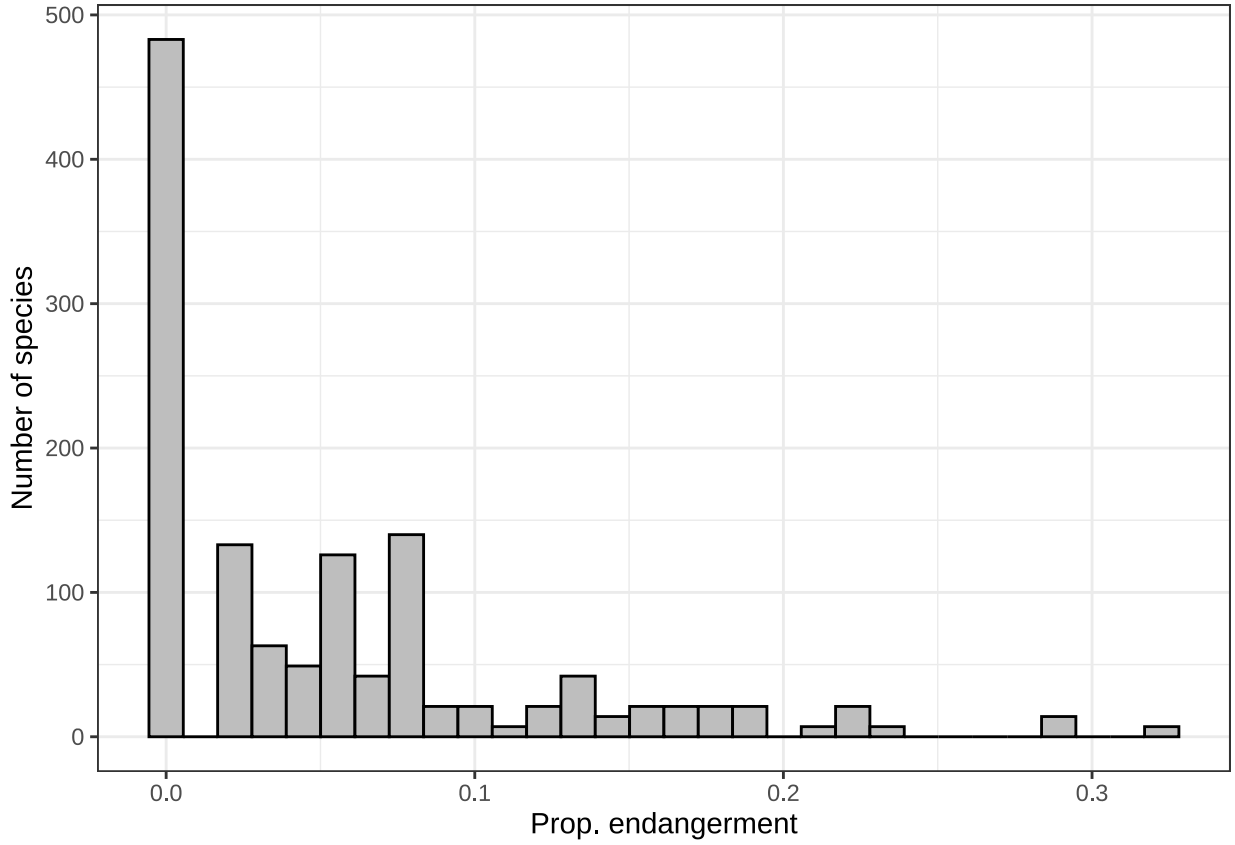
```
## # A tibble: 1,302 x 11
##    species              total_countries threatened_countries perc_endangerment
##    <chr>                          <int>                <int>              <dbl>
##  1 Achillea millefolium              40                    0                  0
##  2 Achillea millefolium              40                    0                  0
##  3 Achillea millefolium              40                    0                  0
##  4 Achillea millefolium              40                    0                  0
```

```
##  5 Achillea millefolium             40                  0             0
##  6 Achillea millefolium             40                  0             0
##  7 Achillea millefolium             40                  0             0
##  8 Aconitum lycoctonum              36                  3          8.33
##  9 Aconitum lycoctonum              36                  3          8.33
## 10 Aconitum lycoctonum              36                  3          8.33
## # i 1,292 more rows
## # i 7 more variables: prop_endangerment <dbl>, stress_tol <dbl>, change <int>,
## #   stage <chr>, mean_slope <dbl>, slope_var <dbl>, sens_01 <fct>
```

Distributions

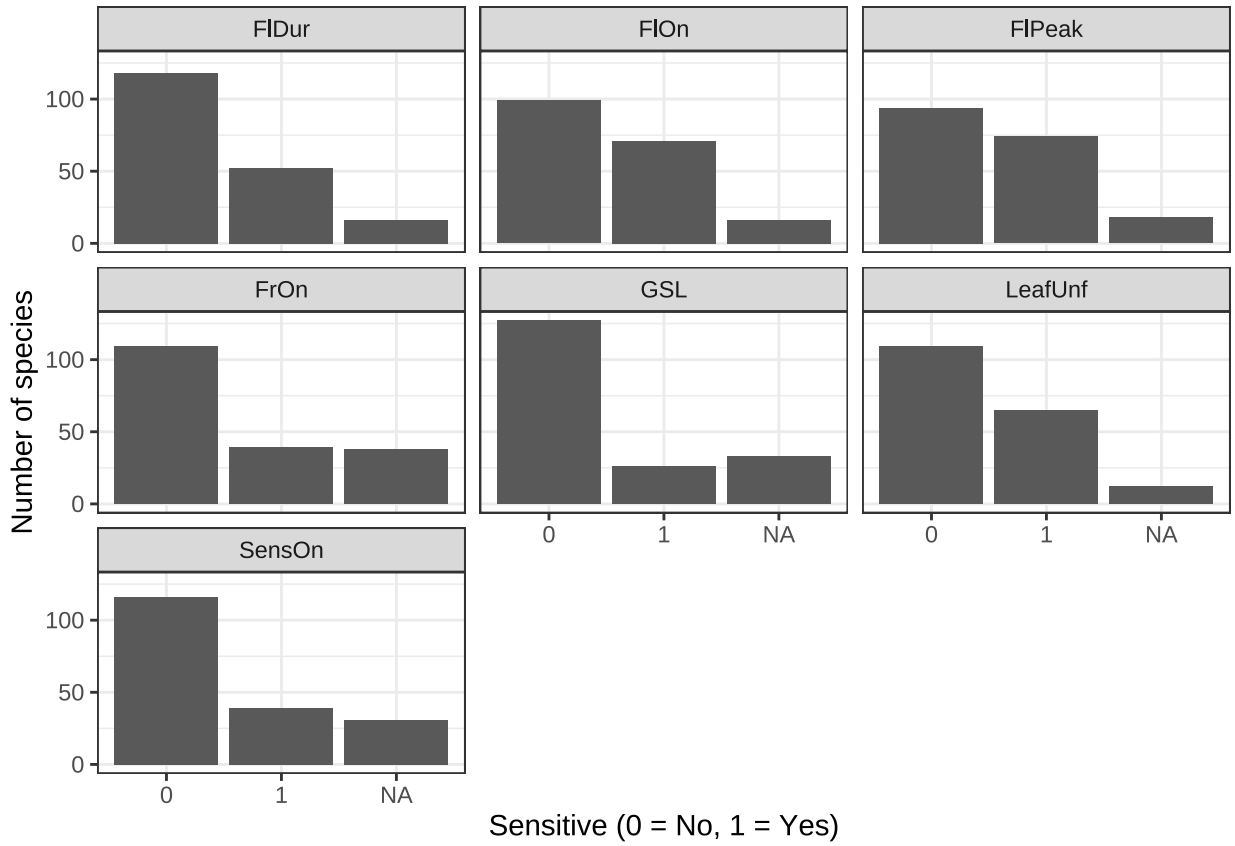
```
ggplot(sens_endanger, aes(x = prop_endangerment)) +
  geom_histogram(color = "black", fill = "grey") +
  labs(x = "Prop. endangerment", y = "Number of species") +
  theme_bw()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
ggplot(sens_endanger, aes(x = factor(sens_01))) +
  geom_bar() +
  facet_wrap(~ stage) +
```

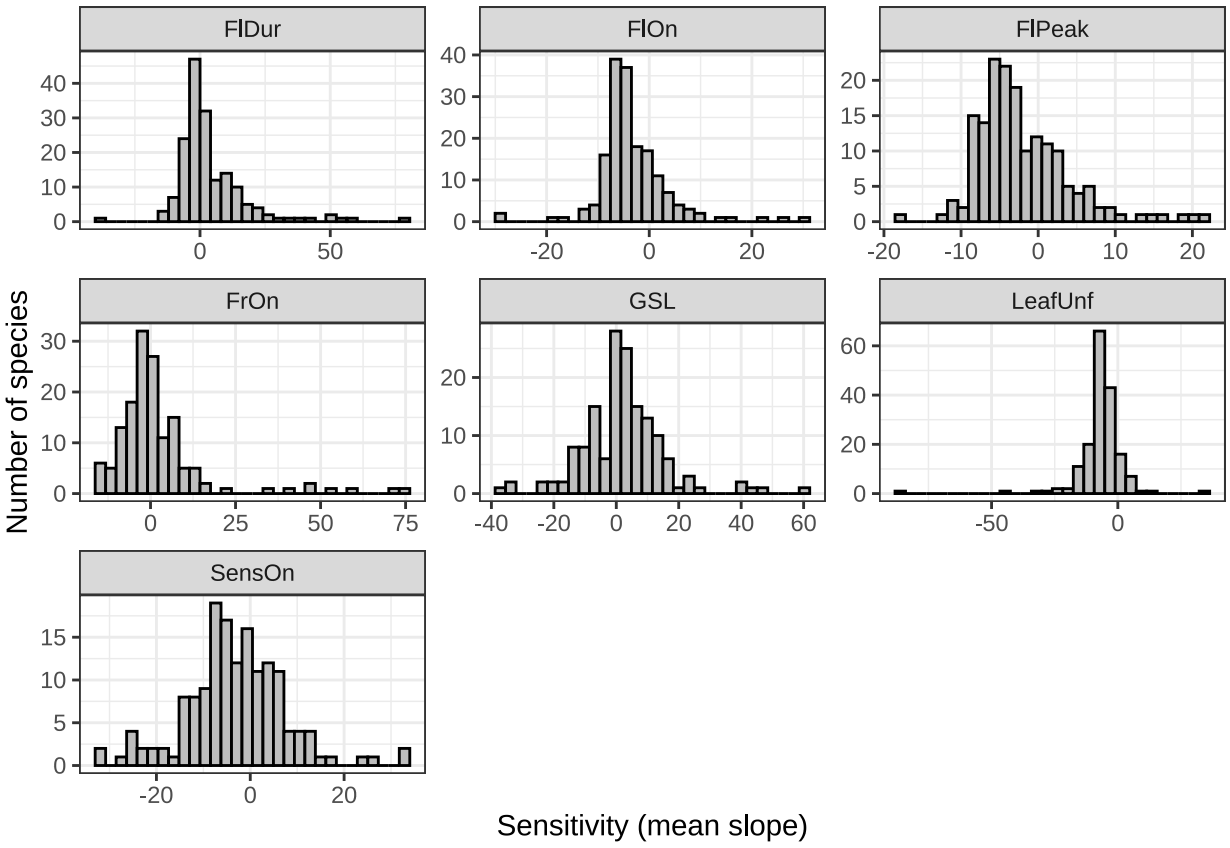
```
  labs(x = "Sensitive (0 = No, 1 = Yes)", y = "Number of species") +
  theme_bw()
```




```
ggplot(sens_endanger, aes(x = mean_slope)) +
  geom_histogram(color = "black", fill = "grey") +
  facet_wrap(~ stage, scales = "free") +
  labs(x = "Sensitivity (mean slope)", y = "Number of species") +
  theme_bw()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
## Warning: Removed 164 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



Models endangerment ~ sensitivity

Filtering slope_var < 50 - clearer patterns.

Binomial models with cbind(threatened_countries, total_countries - threatened_countries).

Added prediction plots only for models with any significant effects.

Sensitive or not

```
stage_names <- sens_endanger %>%
  filter(slope_var < 50) %>%
  distinct(stage) %>%
  pull(stage)

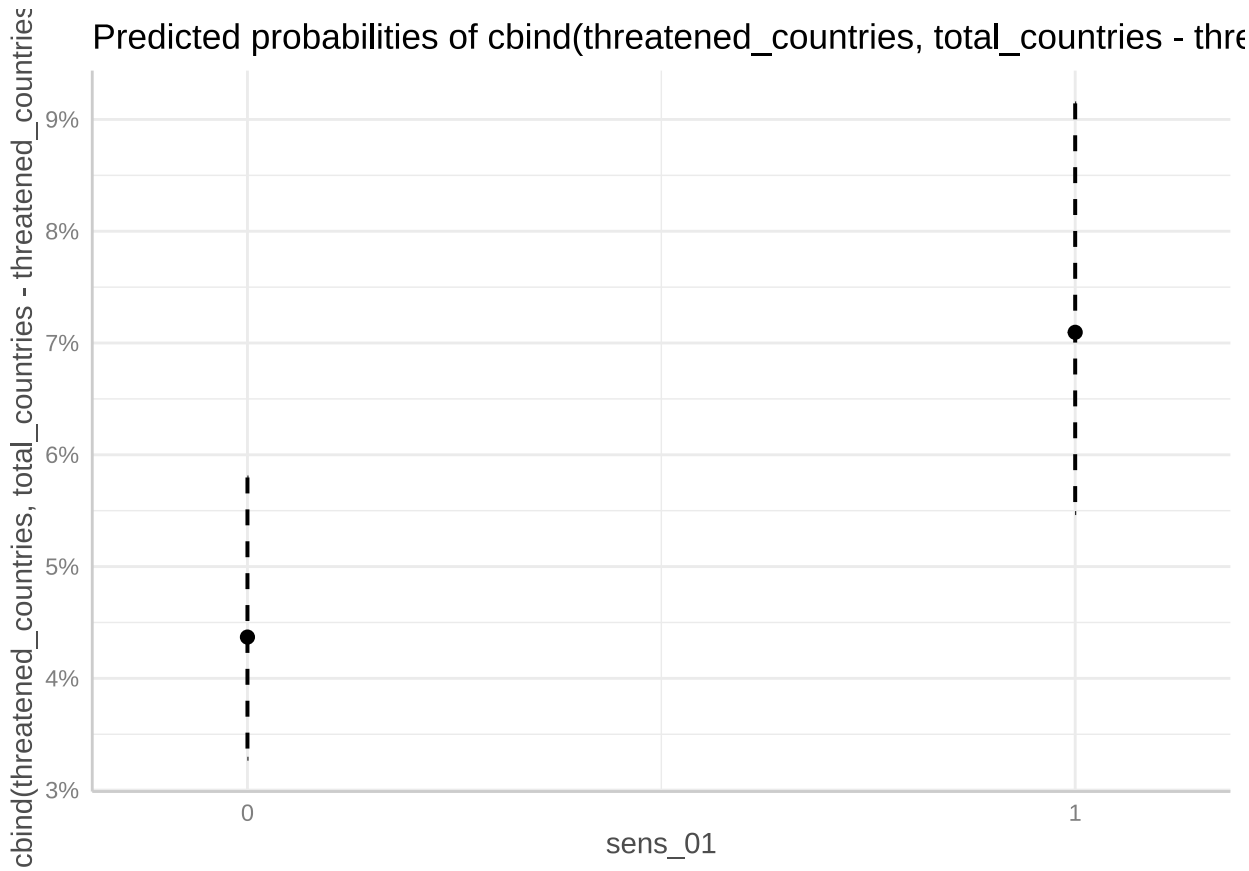
mod_sens_01 <- sens_endanger %>%
  filter(slope_var < 50) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(cbind(threatened_countries,
                            total_countries - threatened_countries) ~ sens_01,
                      data = .x,
                     # Use betabinomial because of overdispersion with binomial
                     family = betabinomial())) %>%
```

```
  setNames(stage_names)

map_df(mod_sens_01, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

```
## # A tibble: 14 x 9
##    stage   effect component term    estimate std.error statistic   p.value sig
##    <chr>   <chr>  <chr>     <chr>      <dbl>     <dbl>     <dbl>     <dbl> <chr>
##  1 FlOn    fixed  cond      (Inter~  -2.85       0.120  -23.6    2.44e-123 "*"
##  2 FlOn    fixed  cond      sens_0~  -0.165      0.222   -0.744  4.57e-  1 ""
##  3 SensOn  fixed  cond      (Inter~  -2.78       0.118  -23.6    1.57e-123 "*"
##  4 SensOn  fixed  cond      sens_0~  -0.221      0.185   -1.20   2.31e-  1 ""
##  5 LeafUnf fixed  cond      (Inter~  -2.78       0.121  -23.0    1.97e-117 "*"
##  6 LeafUnf fixed  cond      sens_0~  -0.278      0.185   -1.50   1.33e-  1 ""
##  7 GSL     fixed  cond      (Inter~  -2.81       0.122  -23.0    3.71e-117 "*"
##  8 GSL     fixed  cond      sens_0~  -0.0656     0.232   -0.282  7.78e-  1 ""
##  9 FlDur   fixed  cond      (Inter~  -2.97       0.185  -16.1    4.93e- 58 "*"
## 10 FlDur   fixed  cond      sens_0~  -0.0861     0.362   -0.238  8.12e-  1 ""
## 11 FlPeak  fixed  cond      (Inter~  -3.09       0.152  -20.3    1.03e- 91 "*"
## 12 FlPeak  fixed  cond      sens_0~   0.514      0.200    2.57   1.02e-  2 "*"
## 13 FrOn    fixed  cond      (Inter~  -2.86       0.165  -17.3    3.63e- 67 "*"
## 14 FrOn    fixed  cond      sens_0~   0.0199     0.264    0.0754 9.40e-  1 ""
```

```
plot(ggpredict(mod_sens_01$FlPeak))
```



Sensitivity sign

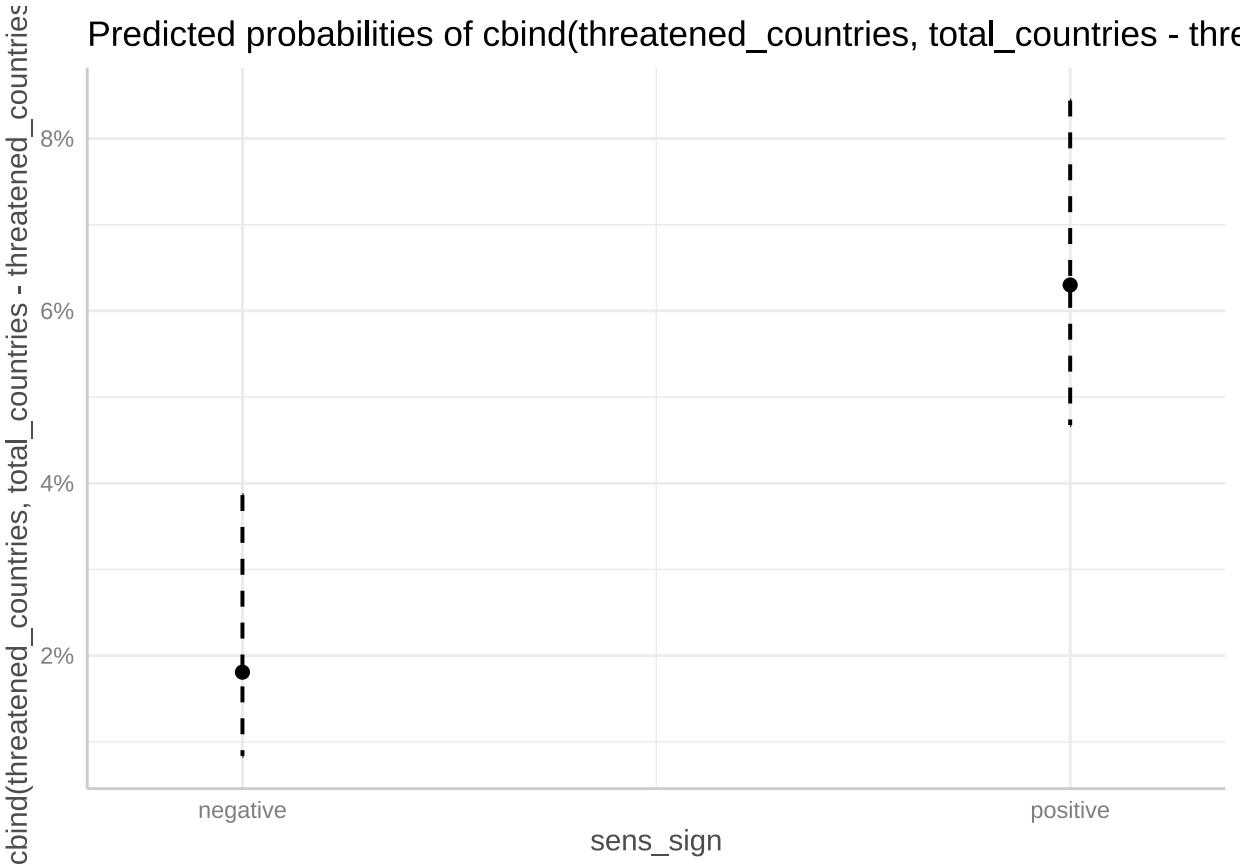
```
sens_endanger <- sens_endanger %>%
  mutate(sens_sign = factor(case_when(
    mean_slope > 0 ~ "positive",
    mean_slope < 0 ~ "negative",
    TRUE ~ NA_character_
  )))
```

```
mod_sens_sign <- sens_endanger %>%
  filter(slope_var < 50) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(cbind(threatened_countries,
                        total_countries - threatened_countries) ~ sens_sign,
                      data = .x, family = betabinomial())) %>%
  setNames(stage_names)

map_df(mod_sens_sign, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

```
## # A tibble: 14 x 9
##    stage   effect component term    estimate std.error statistic   p.value sig
##    <chr>   <chr>  <chr>     <chr>      <dbl>     <dbl>     <dbl>     <dbl> <chr>
##  1 FlOn    fixed  cond      (Inter~  -2.85       0.135  -21.1    4.33e- 99 "*"
##  2 FlOn    fixed  cond      sens_s~  -0.107      0.198   -0.539  5.90e-  1 ""
##  3 SensOn  fixed  cond      (Inter~  -2.83       0.105  -26.9    1.14e-159 "*"
##  4 SensOn  fixed  cond      sens_s~  -0.198      0.238   -0.832  4.06e-  1 ""
##  5 LeafUnf fixed  cond      (Inter~  -2.87       0.111  -25.8    1.65e-146 "*"
##  6 LeafUnf fixed  cond      sens_s~  -0.104      0.208   -0.501  6.17e-  1 ""
##  7 GSL     fixed  cond      (Inter~  -2.82       0.130  -21.7    5.70e-104 "*"
##  8 GSL     fixed  cond      sens_s~  -0.0294     0.211   -0.139  8.89e-  1 ""
##  9 FlDur   fixed  cond      (Inter~  -3.99       0.398  -10.0    1.10e- 23 "*"
## 10 FlDur   fixed  cond      sens_s~   1.30       0.424    3.06   2.24e-  3 "*"
## 11 FlPeak  fixed  cond      (Inter~  -2.78       0.109  -25.4    2.93e-142 "*"
## 12 FlPeak  fixed  cond      sens_s~  -0.780      0.420   -1.86   6.34e-  2 ""
## 13 FrOn    fixed  cond      (Inter~  -2.86       0.157  -18.2    8.28e- 74 "*"
## 14 FrOn    fixed  cond      sens_s~   0.0222     0.277    0.0802 9.36e-  1 ""
```

```
plot(ggpredict(mod_sens_sign$FlDur))
```



Mean slopes

FlDur and GSL: all values

```
stage_names_FlDur_GSL <- sens_endanger %>%
  filter(slope_var < 50) %>%
  filter(stage %in% c("FlDur", "GSL")) %>%
  distinct(stage) %>%
  pull(stage)

mod_sens_all <- sens_endanger %>%
  filter(slope_var < 50) %>%
  filter(stage %in% c("FlDur", "GSL")) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(cbind(threatened_countries,
                            total_countries - threatened_countries) ~ mean_slope,
                      data = .x, family = betabinomial())) %>%
  setNames(stage_names_FlDur_GSL)
map_df(mod_sens_all, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

```
## # A tibble: 4 x 9
##   stage effect component term       estimate std.error statistic   p.value sig
```

```
##   <chr> <chr>  <chr>     <chr>         <dbl>     <dbl>     <dbl>     <dbl> <chr>
## 1 GSL   fixed  cond      (Intercep~ -2.88       0.105    -27.4   7.43e-165 "*"
## 2 GSL   fixed  cond      mean_slope -0.00937    0.0137    -0.684 4.94e-  1 ""
## 3 FlDur fixed  cond      (Intercep~ -3.08       0.180    -17.1   6.81e- 66 "*"
## 4 FlDur fixed  cond      mean_slope  0.0302     0.0201     1.50  1.34e-  1 ""
```

Other stages

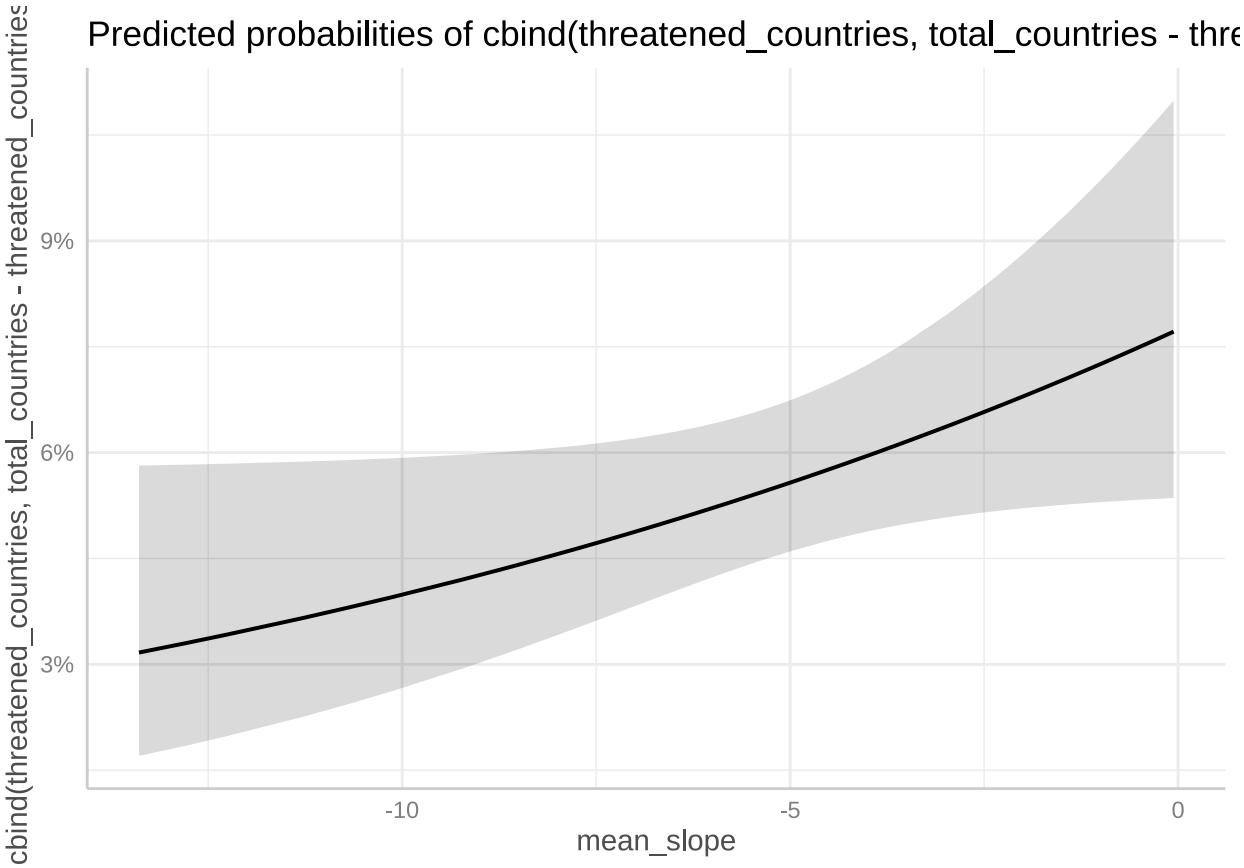
```
stage_names_not_FlDur_GSL <- sens_endanger %>%
  filter(slope_var < 50) %>%
  filter(!stage %in% c("FlDur", "GSL")) %>%
  distinct(stage) %>%
  pull(stage)

mod_sens_neg <- sens_endanger %>%
  filter(slope_var < 50 & mean_slope < 0) %>%
  filter(!stage %in% c("FlDur", "GSL")) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(cbind(threatened_countries,
                        total_countries - threatened_countries) ~ mean_slope,
                  data = .x, family = betabinomial())) %>%
  setNames(stage_names_not_FlDur_GSL)
map_df(mod_sens_neg, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

Negative slopes only

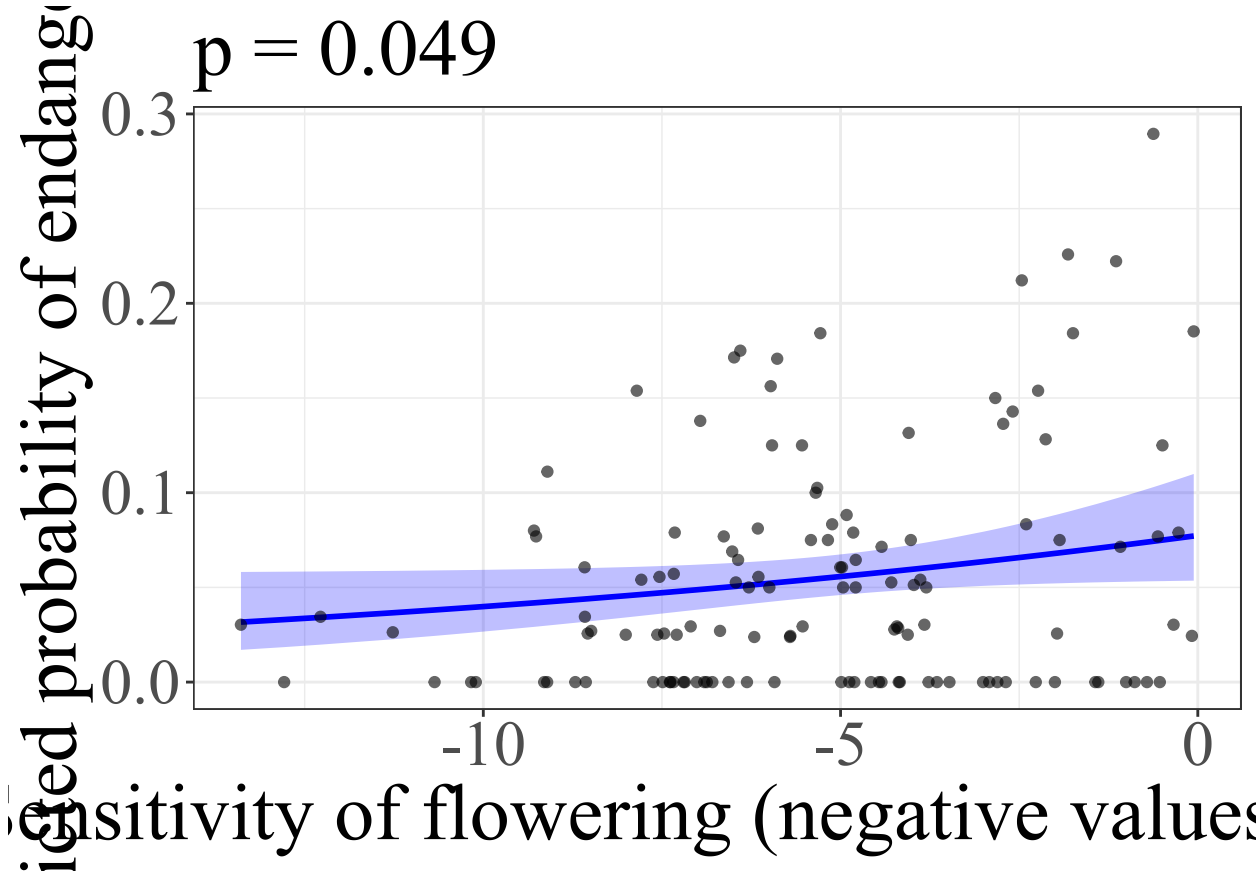
```
## # A tibble: 10 x 9
##    stage   effect component term     estimate std.error statistic  p.value sig
##    <chr>   <chr>  <chr>     <chr>       <dbl>     <dbl>     <dbl>    <dbl> <chr>
##  1 FlOn    fixed  cond      (Interc~ -2.48       0.201    -12.3   5.23e-35 "*"
##  2 FlOn    fixed  cond      mean_sl~  0.0704     0.0358     1.96  4.94e- 2 "*"
##  3 SensOn  fixed  cond      (Interc~ -2.67       0.233    -11.5   2.15e-30 "*"
##  4 SensOn  fixed  cond      mean_sl~  0.0419     0.0435     0.962 3.36e- 1 ""
##  5 LeafUnf fixed  cond      (Interc~ -2.82       0.206    -13.7   1.65e-42 "*"
##  6 LeafUnf fixed  cond      mean_sl~  0.00606    0.0336     0.180 8.57e- 1 ""
##  7 FlPeak  fixed  cond      (Interc~ -3.07       0.213    -14.4   5.77e-47 "*"
##  8 FlPeak  fixed  cond      mean_sl~ -0.0408     0.0237    -1.72  8.53e- 2 ""
##  9 FrOn    fixed  cond      (Interc~ -2.86       0.270    -10.6   2.69e-26 "*"
## 10 FrOn    fixed  cond      mean_sl~ -0.00641    0.0296    -0.216 8.29e- 1 ""
```

```
plot(ggpredict(mod_sens_neg$FlOn, terms="mean_slope [all]")) # p = 0.049
```




```
fig1 <- ggplot() +
    geom_ribbon(data = data.frame(ggpredict(mod_sens_neg$FlOn,
                                            terms="mean_slope [all]")),
                aes(x = x, y = predicted, ymin = conf.low, ymax = conf.high),
                alpha = 0.25, fill = "blue") +
    geom_line(data = data.frame(ggpredict(mod_sens_neg$FlOn,
                                          terms="mean_slope [all]")),
              aes(x = x, y = predicted), color = "blue", linewidth = 1) +
    geom_point(data = sens_endanger %>%
                 filter(mean_slope < 0 & slope_var < 50 & stage == "FlOn"),
               aes(x = mean_slope, y = prop_endangerment), alpha = 0.6) +
    ggtitle("p = 0.049") +
    xlab("Sensitivity of flowering (negative values only)") +
    ylab("Predicted probability of endangerment")
fig1 <- bundle_ggplot(fig1, width = 8, height = 8, units = "cm", scaling = "fixed")
fig1
```

```
## $plot
```




```
##
## $height
## [1] 8
##
## $width
## [1] 8
##
## $units
## [1] "cm"
##
## $bg
## [1] "white"
##
## $scaling
## [1] "fixed"
##
## attr(,"class")
## [1] "bundled_ggplot"
```

```
mod_sens_pos <- sens_endanger %>%
  filter(slope_var < 50 & mean_slope > 0) %>%
  filter(!stage %in% c("FlDur", "GSL")) %>%
  group_by(stage) %>%
```

```
  group_map(~ glmmTMB(cbind(threatened_countries,
                        total_countries - threatened_countries) ~ mean_slope,
                    data = .x, family = betabinomial())) %>%
  setNames(stage_names_not_FlDur_GSL)
map_df(mod_sens_pos, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

Positive slopes only

```
## # A tibble: 10 x 9
##    stage   effect component term     estimate std.error statistic  p.value sig
##    <chr>   <chr>  <chr>     <chr>       <dbl>     <dbl>     <dbl>    <dbl> <chr>
##  1 FlOn    fixed  cond      (Interc~ -2.86       0.311     -9.19  3.79e-20 "*"
##  2 FlOn    fixed  cond      mean_sl~ -0.0369     0.0496    -0.743 4.57e- 1 ""
##  3 SensOn  fixed  cond      (Interc~ -2.74       0.270    -10.1   3.68e-24 "*"
##  4 SensOn  fixed  cond      mean_sl~ -0.0538     0.0486    -1.11  2.68e- 1 ""
##  5 LeafUnf fixed  cond      (Interc~ -2.83       0.239    -11.8   3.63e-32 "*"
##  6 LeafUnf fixed  cond      mean_sl~  0.00539    0.0230     0.234 8.15e- 1 ""
##  7 FlPeak  fixed  cond      (Interc~ -3.70       0.552     -6.71  1.89e-11 "*"
##  8 FlPeak  fixed  cond      mean_sl~ -0.0281     0.110     -0.255 7.99e- 1 ""
##  9 FrOn    fixed  cond      (Interc~ -2.85       0.306     -9.33  1.04e-20 "*"
## 10 FrOn    fixed  cond      mean_sl~ -0.0186     0.0511    -0.365 7.15e- 1 ""
```

Models stress tolerance

Endangerment ~ stress tolerance

```
model_threat_stress <- glmmTMB(cbind(threatened_countries,
                        total_countries - threatened_countries) ~ stress_tol,
                               family = betabinomial(),
                               data = sens_endanger_stress %>%
                          distinct(species,threatened_countries, total_countries, stress_tol))
summary(model_threat_stress)
```

```
##  Family: betabinomial  ( logit )
## Formula:
## cbind(threatened_countries, total_countries - threatened_countries) ~
##     stress_tol
## Data: sens_endanger_stress %>% distinct(species, threatened_countries,
##     total_countries, stress_tol)
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     673.0     682.6    -333.5     667.0       178
##
##
## Dispersion parameter for betabinomial family (): 17.6
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -2.951587   0.147532 -20.006   <2e-16 ***
```

```
## stress_tol   0.005983   0.003750   1.595    0.111
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Endangerment ~ stress tolerance * sensitivity

Filtering slope_var < 50 - clearer patterns.

Sensitive or not

```
mod_sens_01_stress <- sens_endanger_stress %>%
  filter(slope_var < 50) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(cbind(threatened_countries,
                        total_countries - threatened_countries) ~ stress_tol * sens_01,
                      data = .x,
                     # Use betabinomial because of overdispersion with binomial
                     family = betabinomial())) %>%
  setNames(stage_names)

map_df(mod_sens_01_stress, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

```
## # A tibble: 28 x 9
##    stage   effect component term     estimate std.error statistic  p.value sig
##    <chr>   <chr>  <chr>     <chr>       <dbl>     <dbl>     <dbl>    <dbl> <chr>
##  1 FlOn    fixed  cond      (Interc~ -2.91e+0   0.193    -15.1    1.63e-51 "*"
##  2 FlOn    fixed  cond      stress_~  2.48e-3   0.00556    0.445  6.56e- 1 ""
##  3 FlOn    fixed  cond      sens_011 -5.27e-1   0.391     -1.35   1.78e- 1 ""
##  4 FlOn    fixed  cond      stress_~  1.13e-2   0.00891    1.26   2.06e- 1 ""
##  5 SensOn  fixed  cond      (Interc~ -2.86e+0   0.203    -14.1    2.88e-45 "*"
##  6 SensOn  fixed  cond      stress_~  3.02e-3   0.00528    0.572  5.67e- 1 ""
##  7 SensOn  fixed  cond      sens_011 -3.09e-1   0.301     -1.02   3.05e- 1 ""
##  8 SensOn  fixed  cond      stress_~  4.22e-3   0.00817    0.517  6.05e- 1 ""
##  9 LeafUnf fixed  cond      (Interc~ -2.90e+0   0.208    -13.9    4.43e-44 "*"
## 10 LeafUnf fixed  cond      stress_~  5.45e-3   0.00552    0.988  3.23e- 1 ""
## 11 LeafUnf fixed  cond      sens_011 -2.21e-1   0.306     -0.722  4.70e- 1 ""
## 12 LeafUnf fixed  cond      stress_~ -2.83e-3   0.00832   -0.340  7.34e- 1 ""
## 13 GSL     fixed  cond      (Interc~ -3.09e+0   0.220    -14.1    5.06e-45 "*"
## 14 GSL     fixed  cond      stress_~  9.04e-3   0.00569    1.59   1.12e- 1 ""
## 15 GSL     fixed  cond      sens_011  3.29e-1   0.390      0.844  3.99e- 1 ""
## 16 GSL     fixed  cond      stress_~ -9.46e-3   0.00986   -0.960  3.37e- 1 ""
## 17 FlDur   fixed  cond      (Interc~ -3.11e+0   0.259    -12.0    3.33e-33 "*"
## 18 FlDur   fixed  cond      stress_~  6.56e-3   0.00838    0.782  4.34e- 1 ""
## 19 FlDur   fixed  cond      sens_011 -5.12e-1   0.595     -0.860  3.90e- 1 ""
## 20 FlDur   fixed  cond      stress_~  1.98e-2   0.0198     0.997  3.19e- 1 ""
## 21 FlPeak  fixed  cond      (Interc~ -3.12e+0   0.256    -12.2    2.59e-34 "*"
## 22 FlPeak  fixed  cond      stress_~  1.81e-3   0.00745    0.243  8.08e- 1 ""
## 23 FlPeak  fixed  cond      sens_011  3.34e-1   0.335      0.996  3.19e- 1 ""
## 24 FlPeak  fixed  cond      stress_~  5.29e-3   0.00925    0.571  5.68e- 1 ""
## 25 FrOn    fixed  cond      (Interc~ -2.82e+0   0.240    -11.8    5.93e-32 "*"
```

```
## 26 FrOn    fixed  cond      stress_~ -6.85e-4   0.00779   -0.0880 9.30e- 1 ""
## 27 FrOn    fixed  cond      sens_011 -3.78e-1   0.441     -0.858  3.91e- 1 ""
## 28 FrOn    fixed  cond      stress_~  1.40e-2   0.0132     1.06   2.88e- 1 ""
```

Sensitivity sign

```
sens_endanger_stress <- sens_endanger_stress %>%
  mutate(sens_sign = factor(case_when(
    mean_slope > 0 ~ "positive",
    mean_slope < 0 ~ "negative",
    TRUE ~ NA_character_
  )))
```

```
mod_sens_sign_stress <- sens_endanger_stress %>%
  filter(slope_var < 50) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(cbind(threatened_countries,
                      total_countries - threatened_countries) ~ stress_tol * sens_sign,
                    data = .x, family = betabinomial())) %>%
  setNames(stage_names)

map_df(mod_sens_sign_stress, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

```
## # A tibble: 28 x 9
##    stage   effect component term     estimate std.error statistic  p.value sig
##    <chr>   <chr>  <chr>     <chr>       <dbl>     <dbl>     <dbl>    <dbl> <chr>
##  1 FlOn    fixed  cond      (Interc~ -2.94      0.210     -14.0   1.17e-44 "*"
##  2 FlOn    fixed  cond      stress_~  0.00303   0.00579     0.524 6.00e- 1 ""
##  3 FlOn    fixed  cond      sens_si~ -0.281     0.337      -0.834 4.04e- 1 ""
##  4 FlOn    fixed  cond      stress_~  0.00763   0.00860     0.887 3.75e- 1 ""
##  5 SensOn  fixed  cond      (Interc~ -3.05      0.167     -18.3   4.06e-75 "*"
##  6 SensOn  fixed  cond      stress_~  0.00820   0.00418     1.96  4.97e- 2 "*"
##  7 SensOn  fixed  cond      sens_si~  0.363     0.404       0.898 3.69e- 1 ""
##  8 SensOn  fixed  cond      stress_~ -0.0208    0.0127     -1.64  1.00e- 1 ""
##  9 LeafUnf fixed  cond      (Interc~ -2.94      0.178     -16.5   2.80e-61 "*"
## 10 LeafUnf fixed  cond      stress_~  0.00361   0.00501     0.721 4.71e- 1 ""
## 11 LeafUnf fixed  cond      sens_si~ -0.252     0.356      -0.707 4.80e- 1 ""
## 12 LeafUnf fixed  cond      stress_~  0.00352   0.00866     0.406 6.85e- 1 ""
## 13 GSL     fixed  cond      (Interc~ -3.00      0.225     -13.4   9.83e-41 "*"
## 14 GSL     fixed  cond      stress_~  0.00632   0.00554     1.14  2.53e- 1 ""
## 15 GSL     fixed  cond      sens_si~  0.0510    0.376       0.136 8.92e- 1 ""
## 16 GSL     fixed  cond      stress_~ -0.00162   0.00996    -0.162 8.71e- 1 ""
## 17 FlDur   fixed  cond      (Interc~ -4.57      0.796      -5.74  9.63e- 9 "*"
## 18 FlDur   fixed  cond      stress_~  0.0240    0.0270      0.891 3.73e- 1 ""
## 19 FlDur   fixed  cond      sens_si~  1.67      0.819       2.04  4.15e- 2 "*"
## 20 FlDur   fixed  cond      stress_~ -0.0148    0.0277     -0.534 5.93e- 1 ""
## 21 FlPeak  fixed  cond      (Interc~ -2.98      0.179     -16.6   6.39e-62 "*"
## 22 FlPeak  fixed  cond      stress_~  0.00671   0.00465     1.45  1.48e- 1 ""
## 23 FlPeak  fixed  cond      sens_si~ -0.116     0.658      -0.177 8.60e- 1 ""
## 24 FlPeak  fixed  cond      stress_~ -0.0214    0.0223     -0.960 3.37e- 1 ""
## 25 FrOn    fixed  cond      (Interc~ -3.04      0.238     -12.8   2.55e-37 "*"
```

```
## 26 FrOn    fixed  cond      stress_~  0.00800   0.00648     1.24  2.17e- 1 ""
## 27 FrOn    fixed  cond      sens_si~  0.423     0.419       1.01  3.12e- 1 ""
## 28 FrOn    fixed  cond      stress_~ -0.0189    0.0147     -1.28  2.01e- 1 ""
```

Mean slopes

FlDur and GSL: all values

```
mod_sens_all_stress <- sens_endanger_stress %>%
  filter(slope_var < 50) %>%
  filter(stage %in% c("FlDur", "GSL")) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(cbind(threatened_countries,
                        total_countries - threatened_countries) ~ stress_tol * mean_slope,
                    data = .x, family = betabinomial())) %>%
  setNames(stage_names_FlDur_GSL)
map_df(mod_sens_all_stress, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

```
## # A tibble: 8 x 9
##   stage effect component term        estimate std.error statistic  p.value sig
##   <chr> <chr>  <chr>     <chr>          <dbl>     <dbl>     <dbl>    <dbl> <chr>
## 1 GSL   fixed  cond      (Intercept) -3.05e+0  0.168      -18.2   1.18e-73 "*"
## 2 GSL   fixed  cond      stress_tol   5.95e-3  0.00439      1.35  1.76e- 1 ""
## 3 GSL   fixed  cond      mean_slope  -2.09e-2  0.0234      -0.895 3.71e- 1 ""
## 4 GSL   fixed  cond      stress_tol~  5.15e-4  0.000595     0.867 3.86e- 1 ""
## 5 FlDur fixed  cond      (Intercept) -3.25e+0  0.265      -12.3   1.37e-34 "*"
## 6 FlDur fixed  cond      stress_tol   7.54e-3  0.00929      0.812 4.17e- 1 ""
## 7 FlDur fixed  cond      mean_slope   2.11e-2  0.0342       0.617 5.37e- 1 ""
## 8 FlDur fixed  cond      stress_tol~  3.80e-4  0.00127      0.299 7.65e- 1 ""
```

Other stages

```
mod_sens_neg_stress <- sens_endanger_stress %>%
  filter(slope_var < 50 & mean_slope < 0) %>%
  filter(!stage %in% c("FlDur", "GSL")) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(cbind(threatened_countries,
                        total_countries - threatened_countries) ~ stress_tol * mean_slope,
                    data = .x, family = betabinomial())) %>%
  setNames(stage_names_not_FlDur_GSL)
map_df(mod_sens_neg_stress, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

Negative slopes only

```
## # A tibble: 20 x 9
##    stage   effect component term     estimate std.error statistic  p.value sig
```

```
##    <chr>   <chr>  <chr>     <chr>       <dbl>     <dbl>     <dbl>    <dbl> <chr>
##  1 FlOn    fixed  cond      (Interc~ -2.61e+0  0.360      -7.24   4.51e-13 "*"
##  2 FlOn    fixed  cond      stress_~  2.47e-3  0.00800     0.309  7.57e- 1 ""
##  3 FlOn    fixed  cond      mean_sl~  8.02e-2  0.0608      1.32   1.87e- 1 ""
##  4 FlOn    fixed  cond      stress_~ -8.98e-4  0.00146    -0.613  5.40e- 1 ""
##  5 SensOn  fixed  cond      (Interc~ -2.70e+0  0.439      -6.16   7.43e-10 "*"
##  6 SensOn  fixed  cond      stress_~  1.69e-3  0.0106      0.160  8.73e- 1 ""
##  7 SensOn  fixed  cond      mean_sl~  4.19e-2  0.0759      0.552  5.81e- 1 ""
##  8 SensOn  fixed  cond      stress_~ -1.71e-4  0.00193    -0.0886 9.29e- 1 ""
##  9 LeafUnf fixed  cond      (Interc~ -2.91e+0  0.376      -7.75   9.42e-15 "*"
## 10 LeafUnf fixed  cond      stress_~  3.15e-3  0.00873     0.360  7.19e- 1 ""
## 11 LeafUnf fixed  cond      mean_sl~  2.63e-2  0.0625      0.420  6.74e- 1 ""
## 12 LeafUnf fixed  cond      stress_~ -6.79e-4  0.00138    -0.494  6.22e- 1 ""
## 13 FlPeak  fixed  cond      (Interc~ -3.49e+0  0.391      -8.93   4.39e-19 "*"
## 14 FlPeak  fixed  cond      stress_~  1.32e-2  0.00918     1.43   1.51e- 1 ""
## 15 FlPeak  fixed  cond      mean_sl~ -6.89e-2  0.0424     -1.63   1.04e- 1 ""
## 16 FlPeak  fixed  cond      stress_~  8.28e-4  0.000944    0.877  3.81e- 1 ""
## 17 FrOn    fixed  cond      (Interc~ -2.67e+0  0.485      -5.50   3.80e- 8 "*"
## 18 FrOn    fixed  cond      stress_~  1.88e-4  0.0107      0.0176 9.86e- 1 ""
## 19 FrOn    fixed  cond      mean_sl~  6.08e-2  0.0762      0.798  4.25e- 1 ""
## 20 FrOn    fixed  cond      stress_~ -1.45e-3  0.00154    -0.943  3.46e- 1 ""
```

```
mod_sens_pos_stress <- sens_endanger_stress %>%
  filter(slope_var < 50 & mean_slope > 0) %>%
  filter(!stage %in% c("FlDur", "GSL")) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(cbind(threatened_countries,
                            total_countries - threatened_countries) ~ stress_tol * mean_slope,
                      data = .x, family = betabinomial())) %>%
  setNames(stage_names_not_FlDur_GSL)
```

Positive slopes only

```
## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; false convergence (8). See vignette('troubleshooting'),
## help('diagnose')
```

```
map_df(mod_sens_pos_stress, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

```
## # A tibble: 20 x 9
##    stage   effect component term     estimate std.error statistic  p.value sig
##    <chr>   <chr>  <chr>     <chr>       <dbl>     <dbl>     <dbl>    <dbl> <chr>
##  1 FlOn    fixed  cond      (Interc~ -2.82e+0   0.570     -4.95   7.25e- 7 "*"
##  2 FlOn    fixed  cond      stress_~ -1.81e-3   0.0157    -0.115  9.09e- 1 ""
##  3 FlOn    fixed  cond      mean_sl~  7.45e-2   0.110      0.675  5.00e- 1 ""
##  4 FlOn    fixed  cond      stress_~ -5.19e-3   0.00494   -1.05   2.93e- 1 ""
##  5 SensOn  fixed  cond      (Interc~ -3.44e+0   0.429     -8.03   9.92e-16 "*"
##  6 SensOn  fixed  cond      stress_~  2.08e-2   0.00978    2.13   3.33e- 2 "*"
##  7 SensOn  fixed  cond      mean_sl~  9.94e-2   0.0688     1.44   1.49e- 1 ""
```

```
##  8 SensOn  fixed  cond      stress_~ -4.90e-3   0.00240   -2.04   4.09e- 2 "*"
##  9 LeafUnf fixed  cond      (Interc~ -2.86e+0   0.709     -4.03   5.63e- 5 "*"
## 10 LeafUnf fixed  cond      stress_~  3.03e-3   0.0151     0.200  8.41e- 1 ""
## 11 LeafUnf fixed  cond      mean_sl~ -5.99e-3   0.0949    -0.0632 9.50e- 1 ""
## 12 LeafUnf fixed  cond      stress_~  1.58e-4   0.00150    0.105  9.16e- 1 ""
## 13 FlPeak  fixed  cond      (Interc~ -2.72e+0   0.788     -3.45   5.62e- 4 "*"
## 14 FlPeak  fixed  cond      stress_~ -4.34e-2   0.0356    -1.22   2.23e- 1 ""
## 15 FlPeak  fixed  cond      mean_sl~ -1.16e-1   0.136     -0.849  3.96e- 1 ""
## 16 FlPeak  fixed  cond      stress_~  4.58e-3   0.00570    0.804  4.22e- 1 ""
## 17 FrOn    fixed  cond      (Interc~ -2.75e+0   0.442     -6.22   4.86e-10 "*"
## 18 FrOn    fixed  cond      stress_~  1.28e-3   0.0174     0.0733 9.42e- 1 ""
## 19 FrOn    fixed  cond      mean_sl~  1.93e-2   0.0641     0.301  7.64e- 1 ""
## 20 FrOn    fixed  cond      stress_~ -4.14e-3   0.00420   -0.985  3.25e- 1 ""
```

Models Ellenberg tendency of change (raw values) ~ sensitivity

Filtering slope_var < 50 - clearer patterns.

Sensitive or not

```
mod_sens_01_ell <- sens_endanger_stress_ell %>%
  filter(slope_var < 50) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(change ~ sens_01,
                      data = .x,
                     # Use betabinomial because of overdispersion with binomial
                     family = "poisson")) %>%
  setNames(stage_names)

map_df(mod_sens_01_ell, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

```
## # A tibble: 14 x 9
##    stage   effect component term    estimate std.error statistic   p.value sig
##    <chr>   <chr>  <chr>     <chr>      <dbl>     <dbl>     <dbl>     <dbl> <chr>
##  1 FlOn    fixed  cond      (Inter~   1.40      0.0647    21.6   1.00e-103 "*"
##  2 FlOn    fixed  cond      sens_0~  -0.0444    0.143     -0.311 7.56e-  1 ""
##  3 SensOn  fixed  cond      (Inter~   1.36      0.0725    18.7   7.18e- 78 "*"
##  4 SensOn  fixed  cond      sens_0~   0.0837    0.108      0.775 4.38e-  1 ""
##  5 LeafUnf fixed  cond      (Inter~   1.36      0.0765    17.8   1.68e- 70 "*"
##  6 LeafUnf fixed  cond      sens_0~   0.0581    0.108      0.539 5.90e-  1 ""
##  7 GSL     fixed  cond      (Inter~   1.42      0.0702    20.3   3.44e- 91 "*"
##  8 GSL     fixed  cond      sens_0~  -0.0777    0.139     -0.557 5.77e-  1 ""
##  9 FlDur   fixed  cond      (Inter~   1.43      0.0981    14.5   7.01e- 48 "*"
## 10 FlDur   fixed  cond      sens_0~   0.133     0.190      0.700 4.84e-  1 ""
## 11 FlPeak  fixed  cond      (Inter~   1.47      0.0798    18.5   4.91e- 76 "*"
## 12 FlPeak  fixed  cond      sens_0~  -0.129     0.123     -1.05  2.93e-  1 ""
## 13 FrOn    fixed  cond      (Inter~   1.45      0.0884    16.4   1.51e- 60 "*"
## 14 FrOn    fixed  cond      sens_0~  -0.0439    0.168     -0.261 7.94e-  1 ""
```

Sensitivity sign

```
sens_endanger_stress_ell <- sens_endanger_stress_ell %>%
  mutate(sens_sign = factor(case_when(
    mean_slope > 0 ~ "positive",
    mean_slope < 0 ~ "negative",
    TRUE ~ NA_character_
  )))
```

```
mod_sens_sign_ell <- sens_endanger_stress_ell %>%
  filter(slope_var < 50) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(change ~ sens_sign,
                      data = .x, family = "poisson")) %>%
  setNames(stage_names)

map_df(mod_sens_sign_ell, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

```
## # A tibble: 14 x 9
##    stage   effect component term    estimate std.error statistic   p.value sig
##    <chr>   <chr>  <chr>     <chr>      <dbl>     <dbl>     <dbl>     <dbl> <chr>
##  1 FlOn    fixed  cond      (Inter~   1.38      0.0750   18.3    3.53e- 75 "*"
##  2 FlOn    fixed  cond      sens_s~   0.0359    0.117     0.306  7.60e-  1 ""
##  3 SensOn  fixed  cond      (Inter~   1.41      0.0595   23.6    1.46e-123 "*"
##  4 SensOn  fixed  cond      sens_s~  -0.0821    0.138    -0.593  5.53e-  1 ""
##  5 LeafUnf fixed  cond      (Inter~   1.39      0.0618   22.6    9.97e-113 "*"
##  6 LeafUnf fixed  cond      sens_s~  -0.0318    0.127    -0.251  8.02e-  1 ""
##  7 GSL     fixed  cond      (Inter~   1.40      0.0733   19.1    6.09e- 81 "*"
##  8 GSL     fixed  cond      sens_s~   0.0127    0.130     0.0975 9.22e-  1 ""
##  9 FlDur   fixed  cond      (Inter~   1.56      0.132    11.8    6.00e- 32 "*"
## 10 FlDur   fixed  cond      sens_s~  -0.160     0.171    -0.935  3.50e-  1 ""
## 11 FlPeak  fixed  cond      (Inter~   1.41      0.0639   22.0    1.79e-107 "*"
## 12 FlPeak  fixed  cond      sens_s~   0.0972    0.203     0.479  6.32e-  1 ""
## 13 FrOn    fixed  cond      (Inter~   1.46      0.0913   15.9    3.24e- 57 "*"
## 14 FrOn    fixed  cond      sens_s~  -0.0513    0.161    -0.319  7.50e-  1 ""
```

Mean slopes

FlDur and GSL: all values

```
mod_sens_all_ell <- sens_endanger_stress_ell %>%
  filter(slope_var < 50) %>%
  filter(stage %in% c("FlDur", "GSL")) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(change ~ mean_slope,
                  data = .x, family = "poisson")) %>%
  setNames(stage_names_FlDur_GSL)
map_df(mod_sens_all_ell, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

```
## # A tibble: 4 x 9
##   stage effect component term       estimate std.error statistic   p.value sig
##   <chr> <chr>  <chr>     <chr>         <dbl>     <dbl>     <dbl>     <dbl> <chr>
## 1 GSL   fixed  cond      (Intercep~  1.39      0.0578     24.0   1.54e-127 "*"
## 2 GSL   fixed  cond      mean_slope  0.00732   0.00865     0.847 3.97e-  1 ""
## 3 FlDur fixed  cond      (Intercep~  1.47      0.0841     17.5   3.29e- 68 "*"
## 4 FlDur fixed  cond      mean_slope -0.00995   0.0104     -0.958 3.38e-  1 ""
```

Other stages

```
mod_sens_neg_ell <- sens_endanger_stress_ell %>%
  filter(slope_var < 50 & mean_slope < 0) %>%
  filter(!stage %in% c("FlDur", "GSL")) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(change ~ mean_slope,
                  data = .x, family = "poisson")) %>%
  setNames(stage_names_not_FlDur_GSL)
map_df(mod_sens_neg_ell, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

Negative slopes only

```
## # A tibble: 10 x 9
##    stage   effect component term     estimate std.error statistic  p.value sig
##    <chr>   <chr>  <chr>     <chr>       <dbl>     <dbl>     <dbl>    <dbl> <chr>
##  1 FlOn    fixed  cond      (Interc~  1.31       0.135      9.73  2.23e-22 "*"
##  2 FlOn    fixed  cond      mean_sl~ -0.0182     0.0221    -0.823 4.11e- 1 ""
##  3 SensOn  fixed  cond      (Interc~  1.47       0.120     12.2   2.90e-34 "*"
##  4 SensOn  fixed  cond      mean_sl~  0.0143     0.0203     0.706 4.80e- 1 ""
##  5 LeafUnf fixed  cond      (Interc~  1.45       0.125     11.6   6.66e-31 "*"
##  6 LeafUnf fixed  cond      mean_sl~  0.00968    0.0198     0.488 6.25e- 1 ""
##  7 FlPeak  fixed  cond      (Interc~  1.48       0.126     11.7   7.94e-32 "*"
##  8 FlPeak  fixed  cond      mean_sl~  0.0107     0.0160     0.666 5.05e- 1 ""
##  9 FrOn    fixed  cond      (Interc~  1.43       0.142     10.1   9.09e-24 "*"
## 10 FrOn    fixed  cond      mean_sl~ -0.00345    0.0140    -0.246 8.05e- 1 ""
```

```
mod_sens_pos_ell <- sens_endanger_stress_ell %>%
  filter(slope_var < 50 & mean_slope > 0) %>%
  filter(!stage %in% c("FlDur", "GSL")) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(change ~ mean_slope,
                  data = .x, family = "poisson")) %>%
  setNames(stage_names_not_FlDur_GSL)
map_df(mod_sens_pos_ell, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

Positive slopes only

```
## # A tibble: 10 x 9
##    stage   effect component term     estimate std.error statistic  p.value sig
##    <chr>   <chr>  <chr>     <chr>       <dbl>     <dbl>     <dbl>    <dbl> <chr>
##  1 FlOn    fixed  cond      (Interc~   1.48      0.202      7.32  2.52e-13 "*"
##  2 FlOn    fixed  cond      mean_sl~  -0.0371    0.0407    -0.911 3.62e- 1 ""
##  3 SensOn  fixed  cond      (Interc~   1.53      0.160      9.59  8.80e-22 "*"
##  4 SensOn  fixed  cond      mean_sl~  -0.0499    0.0370    -1.35  1.78e- 1 ""
##  5 LeafUnf fixed  cond      (Interc~   1.52      0.164      9.30  1.42e-20 "*"
##  6 LeafUnf fixed  cond      mean_sl~  -0.0247    0.0283    -0.873 3.83e- 1 ""
##  7 FlPeak  fixed  cond      (Interc~   1.56      0.286      5.46  4.82e- 8 "*"
##  8 FlPeak  fixed  cond      mean_sl~  -0.0122    0.0455    -0.268 7.88e- 1 ""
##  9 FrOn    fixed  cond      (Interc~   1.50      0.220      6.82  9.10e-12 "*"
## 10 FrOn    fixed  cond      mean_sl~  -0.0308    0.0567    -0.543 5.87e- 1 ""
```

Models Ellenberg tendency of change (declining / not) ~ sensitivity

Declining species defined as those with a trend index <= 3 (as in Ozinga et al. 2009).

We create a new variable "decline", which is 1 if trend index <= 3, and 0 otherwise.

```
sens_endanger_stress_ell <- sens_endanger_stress_ell %>%
  mutate(decline = ifelse(is.na(change), NA, ifelse(change <= 3, 1, 0)))
```

Filtering slope_var < 50 - clearer patterns.

Sensitive or not

```
mod_sens_01_decl <- sens_endanger_stress_ell %>%
  filter(slope_var < 50) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(decline ~ sens_01,
                      data = .x,
                     family = binomial)) %>%
  setNames(stage_names)

map_df(mod_sens_01_decl, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

```
## # A tibble: 14 x 9
##    stage   effect component term      estimate std.error statistic p.value sig
##    <chr>   <chr>  <chr>     <chr>        <dbl>     <dbl>     <dbl>   <dbl> <chr>
##  1 FlOn    fixed  cond      (Interce~   -0.593     0.272    -2.18  2.92e-2 "*"
##  2 FlOn    fixed  cond      sens_011    -0.195     0.604    -0.323 7.46e-1 ""
##  3 SensOn  fixed  cond      (Interce~   -0.544     0.296    -1.83  6.66e-2 ""
##  4 SensOn  fixed  cond      sens_011    -0.317     0.466    -0.679 4.97e-1 ""
##  5 LeafUnf fixed  cond      (Interce~   -0.560     0.313    -1.79  7.42e-2 ""
##  6 LeafUnf fixed  cond      sens_011    -0.243     0.458    -0.530 5.96e-1 ""
##  7 GSL     fixed  cond      (Interce~   -1.13      0.332    -3.39  7.00e-4 "*"
##  8 GSL     fixed  cond      sens_011     0.903     0.579     1.56  1.19e-1 ""
##  9 FlDur   fixed  cond      (Interce~   -0.754     0.429    -1.76  7.87e-2 ""
```

```
## 10 FlDur   fixed  cond      sens_011    -1.19      1.15     -1.04  3.01e-1 ""
## 11 FlPeak  fixed  cond      (Interce~   -1.10      0.385    -2.85  4.31e-3 "*"
## 12 FlPeak  fixed  cond      sens_011     0.552     0.540     1.02  3.07e-1 ""
## 13 FrOn    fixed  cond      (Interce~   -1.19      0.432    -2.76  5.85e-3 "*"
## 14 FrOn    fixed  cond      sens_011     0.496     0.749     0.663 5.08e-1 ""
```

Sensitivity sign

```
mod_sens_sign_decl <- sens_endanger_stress_ell %>%
  filter(slope_var < 50) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(decline ~ sens_sign,
                      data = .x, family = binomial)) %>%
  setNames(stage_names)

map_df(mod_sens_sign_decl, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

```
## # A tibble: 14 x 9
##    stage   effect component term      estimate std.error statistic p.value sig
##    <chr>   <chr>  <chr>     <chr>        <dbl>     <dbl>     <dbl>   <dbl> <chr>
##  1 FlOn    fixed  cond      (Interce~  -0.595      0.311    -1.91  0.0562  ""
##  2 FlOn    fixed  cond      sens_sig~  -0.0984     0.497    -0.198 0.843   ""
##  3 SensOn  fixed  cond      (Interce~  -0.759      0.258    -2.94  0.00330 "*"
##  4 SensOn  fixed  cond      sens_sig~   0.402      0.556     0.723 0.470   ""
##  5 LeafUnf fixed  cond      (Interce~  -0.740      0.265    -2.79  0.00529 "*"
##  6 LeafUnf fixed  cond      sens_sig~   0.254      0.522     0.487 0.626   ""
##  7 GSL     fixed  cond      (Interce~  -0.629      0.310    -2.03  0.0423  "*"
##  8 GSL     fixed  cond      sens_sig~  -0.818      0.636    -1.29  0.198   ""
##  9 FlDur   fixed  cond      (Interce~  -1.61       0.775    -2.08  0.0377  "*"
## 10 FlDur   fixed  cond      sens_sig~   0.916      0.902     1.02  0.310   ""
## 11 FlPeak  fixed  cond      (Interce~  -0.769      0.278    -2.77  0.00558 "*"
## 12 FlPeak  fixed  cond      sens_sig~  -0.840      1.13     -0.744 0.457   ""
## 13 FrOn    fixed  cond      (Interce~  -0.916      0.418    -2.19  0.0285  "*"
## 14 FrOn    fixed  cond      sens_sig~  -0.383      0.774    -0.495 0.621   ""
```

Mean slopes

FlDur and GSL: all values

```
mod_sens_all_decl <- sens_endanger_stress_ell %>%
  filter(slope_var < 50) %>%
  filter(stage %in% c("FlDur", "GSL")) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(decline ~ mean_slope,
                  data = .x, family = binomial)) %>%
  setNames(stage_names_FlDur_GSL)
map_df(mod_sens_all_decl, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

```
## # A tibble: 4 x 9
##   stage effect component term        estimate std.error statistic p.value sig
##   <chr> <chr>  <chr>     <chr>          <dbl>     <dbl>     <dbl>   <dbl> <chr>
## 1 GSL   fixed  cond      (Intercept)  -0.637     0.244     -2.61  0.00904 "*"
## 2 GSL   fixed  cond      mean_slope   -0.0330    0.0404    -0.816 0.414   ""
## 3 FlDur fixed  cond      (Intercept)  -1.11      0.434     -2.56  0.0104  "*"
## 4 FlDur fixed  cond      mean_slope    0.0643    0.0604     1.07  0.287   ""
```

Other stages

```
mod_sens_neg_decl <- sens_endanger_stress_ell %>%
  filter(slope_var < 50 & mean_slope < 0) %>%
  filter(!stage %in% c("FlDur", "GSL")) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(decline ~ mean_slope,
                  data = .x, family = binomial)) %>%
  setNames(stage_names_not_FlDur_GSL)
map_df(mod_sens_neg_decl, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

Negative slopes only

```
## # A tibble: 10 x 9
##    stage   effect component term      estimate std.error statistic p.value sig
##    <chr>   <chr>  <chr>     <chr>        <dbl>     <dbl>     <dbl>   <dbl> <chr>
##  1 FlOn    fixed  cond      (Interce~  -0.316     0.568     -0.556  0.578  ""
##  2 FlOn    fixed  cond      mean_slo~   0.0848    0.0988     0.859  0.391  ""
##  3 SensOn  fixed  cond      (Interce~  -1.26      0.540     -2.32   0.0201 "*"
##  4 SensOn  fixed  cond      mean_slo~  -0.0962    0.0856    -1.12   0.261  ""
##  5 LeafUnf fixed  cond      (Interce~  -1.37      0.579     -2.37   0.0178 "*"
##  6 LeafUnf fixed  cond      mean_slo~  -0.135     0.0851    -1.59   0.112  ""
##  7 FlPeak  fixed  cond      (Interce~  -1.25      0.565     -2.21   0.0271 "*"
##  8 FlPeak  fixed  cond      mean_slo~  -0.0671    0.0673    -0.997  0.319  ""
##  9 FrOn    fixed  cond      (Interce~  -0.402     0.682     -0.589  0.556  ""
## 10 FrOn    fixed  cond      mean_slo~   0.0738    0.0842     0.876  0.381  ""
```

```
mod_sens_pos_decl <- sens_endanger_stress_ell %>%
  filter(slope_var < 50 & mean_slope > 0) %>%
  filter(!stage %in% c("FlDur", "GSL")) %>%
  group_by(stage) %>%
  group_map(~ glmmTMB(decline ~ mean_slope,
                  data = .x, family = binomial)) %>%
  setNames(stage_names_not_FlDur_GSL)
map_df(mod_sens_pos_decl, tidy, .id = "stage") %>%
  mutate(sig = ifelse(p.value < 0.05, "*", "")) %>% print(n = 50)
```

Positive slopes only

```
## # A tibble: 10 x 9
##    stage   effect component term      estimate std.error statistic p.value sig
##    <chr>   <chr>  <chr>     <chr>        <dbl>     <dbl>     <dbl>   <dbl> <chr>
##  1 FlOn    fixed  cond      (Interce~ -1.34        0.921  -1.46     0.146  ""
##  2 FlOn    fixed  cond      mean_slo~  0.223       0.168   1.33     0.184  ""
##  3 SensOn  fixed  cond      (Interce~ -1.24        0.751  -1.65     0.0987 ""
##  4 SensOn  fixed  cond      mean_slo~  0.205       0.165   1.25     0.212  ""
##  5 LeafUnf fixed  cond      (Interce~ -2.46        1.02   -2.42     0.0154 "*"
##  6 LeafUnf fixed  cond      mean_slo~  0.182       0.134   1.36     0.174  ""
##  7 FlPeak  fixed  cond      (Interce~ -1.58        1.64   -0.964    0.335  ""
##  8 FlPeak  fixed  cond      mean_slo~ -0.00540     0.254  -0.0212   0.983  ""
##  9 FrOn    fixed  cond      (Interce~ -1.31        1.10   -1.19     0.235  ""
## 10 FrOn    fixed  cond      mean_slo~  0.00188     0.270   0.00695  0.994  ""
```

PCA

```
pca_phen <- prcomp(sens_per_endanger %>%
                     select(starts_with("mean_slope_")) %>%
                     drop_na(),
                   center = TRUE, scale. = TRUE)
```

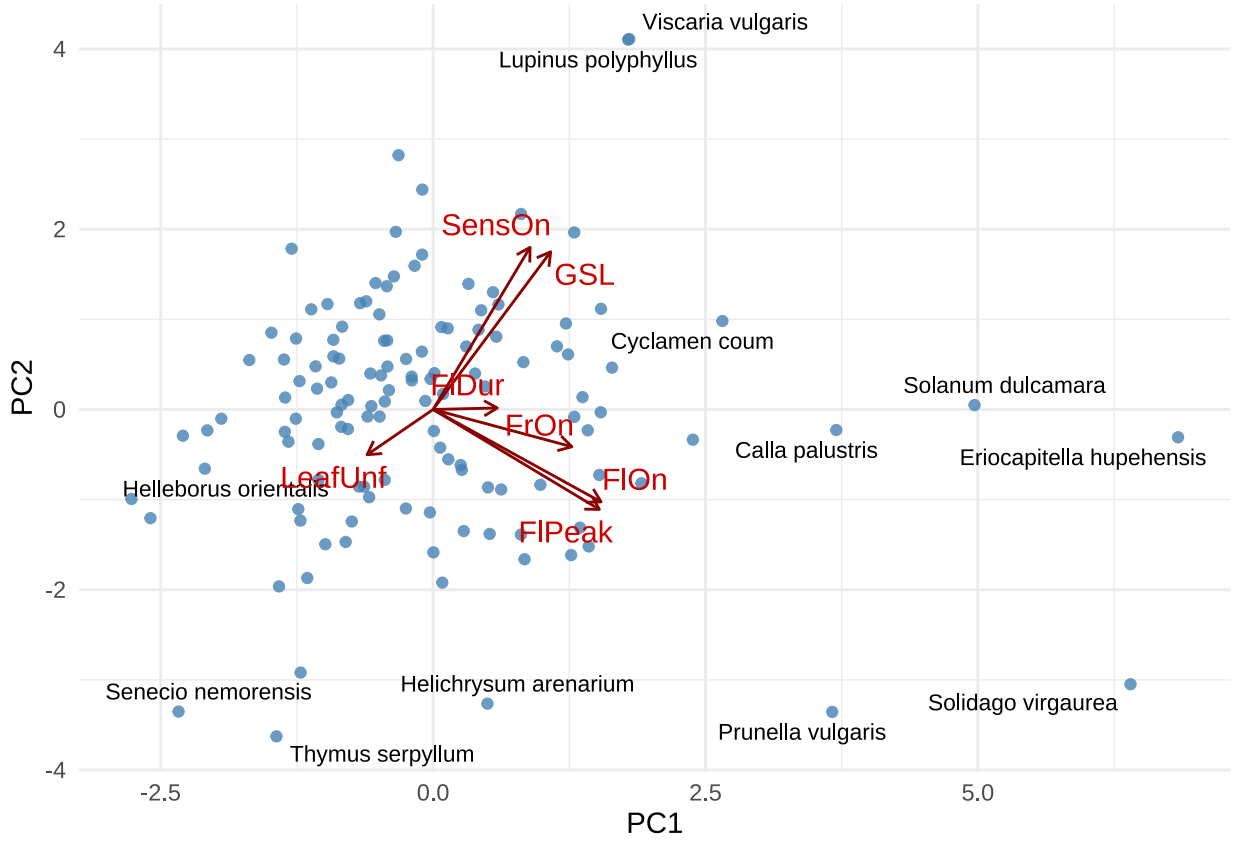
```
scores <- as.data.frame(pca_phen$x) %>%
  mutate(species = (sens_per_endanger %>%
                      select(species, starts_with("mean_slope_")) %>%
                      drop_na())$species) %>%
  mutate(label = ifelse(abs(PC1) > quantile(abs(PC1), 0.95) |
                          abs(PC2) > quantile(abs(PC2), 0.95),
                        species, ""))
```

```
# Variable loadings for arrows
loadings <- as.data.frame(pca_phen$rotation) %>%
  mutate(variable = rownames(.)) %>%
  # Remove mean_slope from variable names
  mutate(variable = gsub("mean_slope_", "", variable))

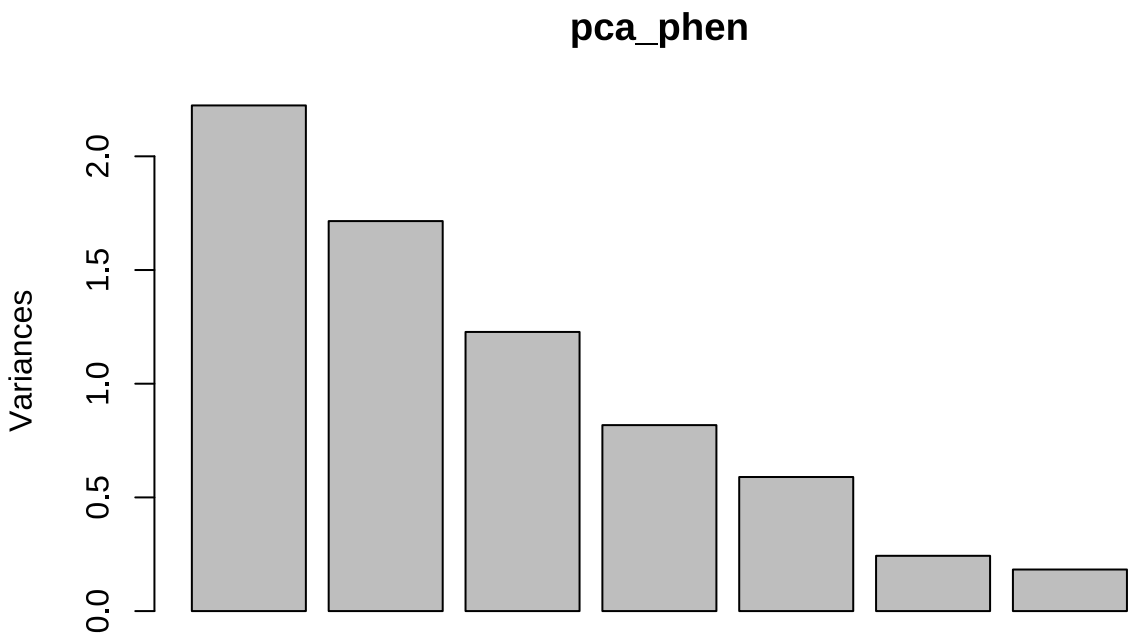
ggplot(scores %>%
         left_join(
           sens_per_endanger %>%
             select(species, prop_endangerment)
           ),
       aes(x = PC1, y = PC2)) +
  geom_point(alpha = 0.8, color = "steelblue"
             ) +
  geom_text_repel(aes(label = label), size = 3, max.overlaps = 15) +
  geom_segment(data = loadings,
               aes(x = 0, y = 0, xend = PC1*3, yend = PC2*3),
               arrow = arrow(length = unit(0.2, "cm")),
               color = "darkred") +
  geom_text_repel(data = loadings,
                  aes(x = PC1*3, y = PC2*3, label = variable),
                  color = "red3", size = 4) +
```

```
  theme_minimal() +
  labs(x = "PC1", y = "PC2")
```

```
## Joining with `by = join_by(species)`
```

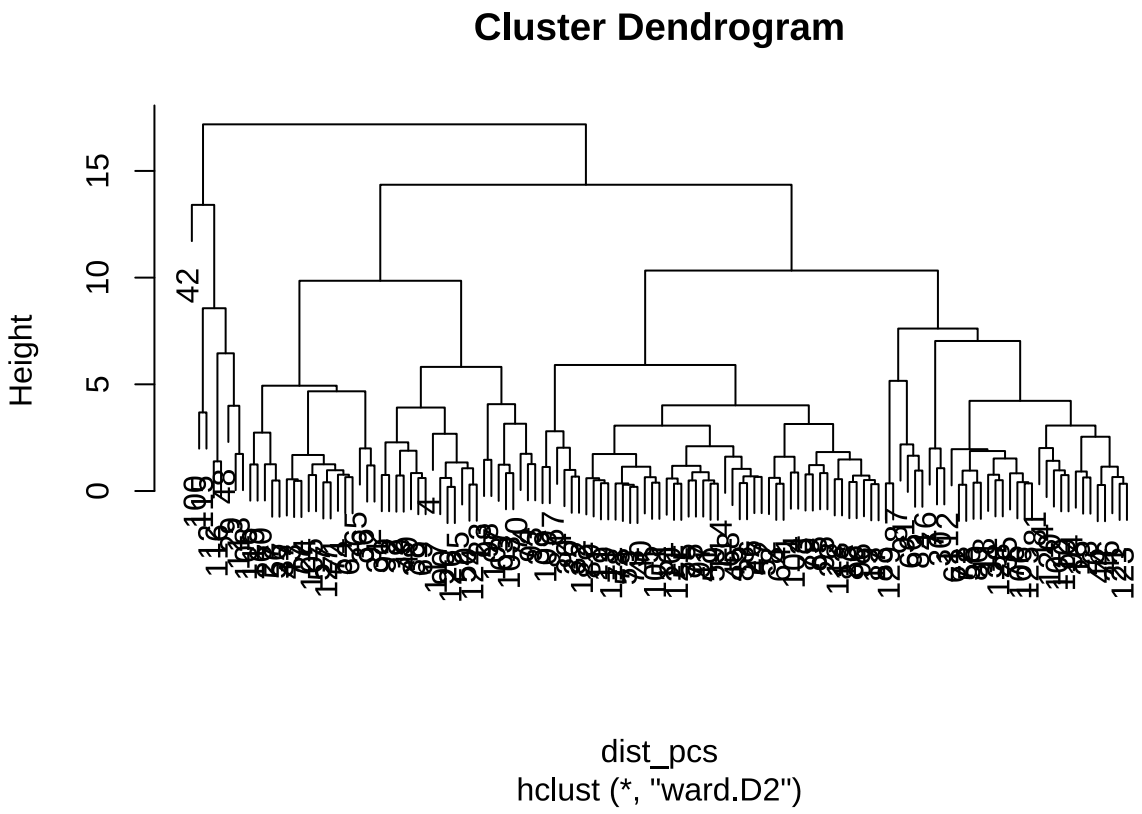



```
plot(pca_phen)
```

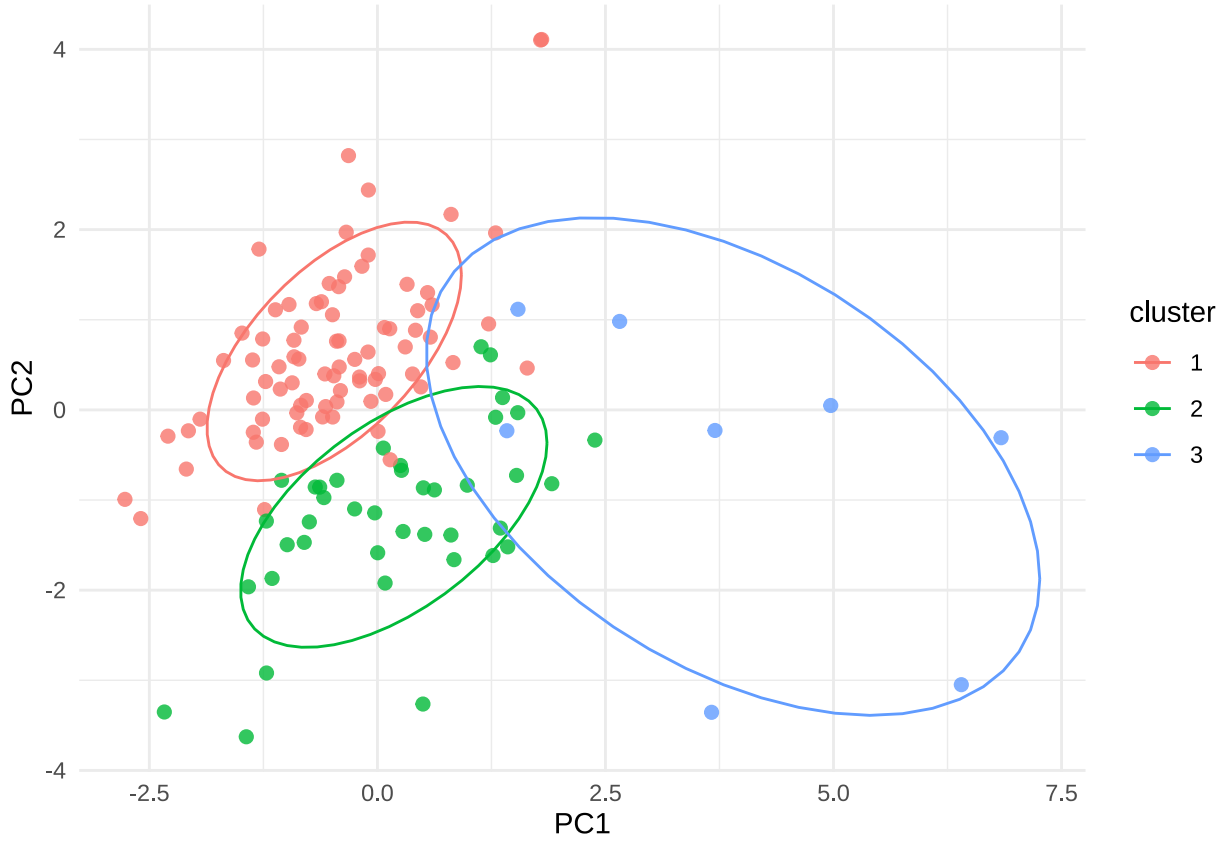


Clustering

```
dist_pcs <- dist(scores %>%
                   select(PC1, PC2, PC3, PC4),
                 method = "euclidean")
hc <- hclust(dist_pcs, method = "ward.D2")
plot(hc)
```



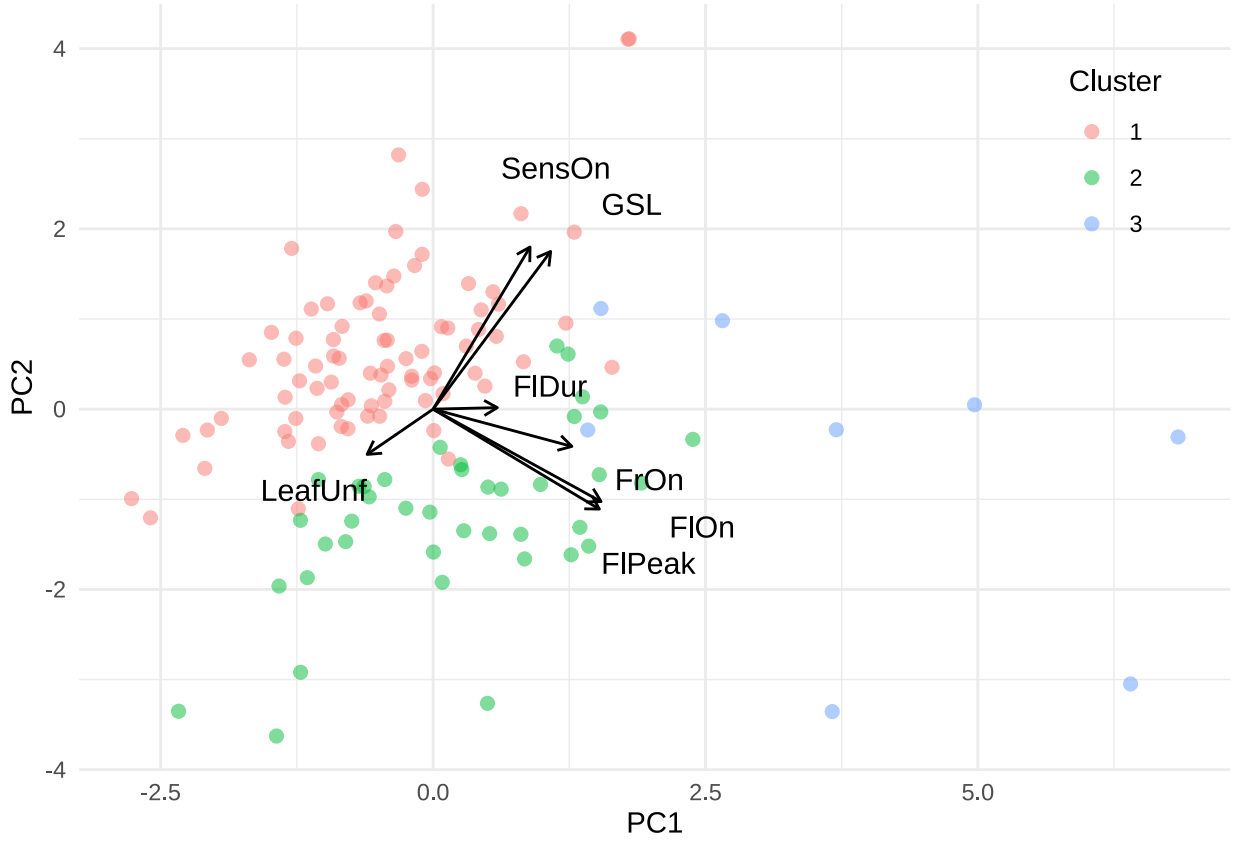

```
scores$cluster <- factor(cutree(hc, k = 3))
ggplot(scores, aes(PC1, PC2, color = cluster)) +
  geom_point(size = 2, alpha = 0.8) +
  stat_ellipse(type = "norm", level = 0.68) +
  theme_minimal()
```



Plot with clusters

```
ggplot(
  scores %>%
    left_join(
      sens_per_endanger %>%
        select(species, prop_endangerment),
      by = "species"
    ),
  aes(x = PC1, y = PC2, color = cluster)
) +
  geom_point(alpha = 0.5, size = 2) +
  #geom_text_repel(aes(label = label), size = 3, max.overlaps = 15) +
  geom_segment(
    data = loadings,
    aes(x = 0, y = 0, xend = PC1 * 3, yend = PC2 * 3),
    arrow = arrow(length = unit(0.2, "cm")),
    color = "black",
    inherit.aes = FALSE
  ) +
  geom_text_repel(
    data = loadings,
    aes(x = PC1 * 4, y = PC2 * 4, label = variable),
    color = "black",
```

```
  size = 4,
  inherit.aes = FALSE,
  nudge_x = 0.1* sign(loadings$PC1),
  nudge_y = 0.1 * sign(loadings$PC2),
  box.padding = 0.2,      # distance around text box
  point.padding = 0.2,    # distance from arrows/points
  max.overlaps = Inf
  ) +
theme_minimal() +
labs(
  x = "PC1",
  y = "PC2",
  color = "Cluster"
) +
#stat_ellipse(type = "norm", level = 0.68) +
theme(legend.position = c(0.9, 0.8))
```



RDA effect of endangerment

Endangered or not

```
rda_trait_endanger <- rda(
  sens_per_endanger %>% select(starts_with("mean_slope_")) %>% drop_na() ~
    (sens_per_endanger %>%
       mutate(endangerment = factor(
         ifelse(prop_endangerment == 0, 0, 1)))%>% drop_na())$endangerment,
  data = sens_per_endanger
  )
anova(rda_trait_endanger, permutations = 999)
```

```
## Permutation test for rda under reduced model
## Permutation: free
## Number of permutations: 999
##
## Model: rda(formula = sens_per_endanger %>% select(starts_with("mean_slope_")) %>% drop_na() ~ (sens_p
##           Df Variance      F Pr(>F)
## Model      1    25.42 3.9627  0.002 **
## Residual 127   814.81
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(rda_trait_endanger, by = "axis", permutations = 999)
```

```
## Permutation test for rda under reduced model
## Forward tests for axes
## Permutation: free
## Number of permutations: 999
##
## Model: rda(formula = sens_per_endanger %>% select(starts_with("mean_slope_")) %>% drop_na() ~ (sens_p
##           Df Variance      F Pr(>F)
## RDA1       1    25.42 3.9627  0.006 **
## Residual 127   814.81
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
RsquareAdj(rda_trait_endanger)
```

```
## $r.squared
## [1] 0.03025808
##
## $adj.r.squared
## [1] 0.02262231
```

Proportion of endangerment

```
rda_trait_endanger_prop <- rda(
  sens_per_endanger %>% select(starts_with("mean_slope_")) %>% drop_na() ~
    (sens_per_endanger%>% drop_na())$prop_endangerment,
  data = sens_per_endanger
  )
anova(rda_trait_endanger_prop, permutations = 999)
```

```
## Permutation test for rda under reduced model
## Permutation: free
## Number of permutations: 999
##
## Model: rda(formula = sens_per_endanger %>% select(starts_with("mean_slope_")) %>% drop_na() ~ (sens_
##           Df Variance      F Pr(>F)
## Model      1    12.31 1.8879   0.11
## Residual 127   827.93
```

```
anova(rda_trait_endanger_prop, by = "axis", permutations = 999)
```

```
## Permutation test for rda under reduced model
## Forward tests for axes
## Permutation: free
## Number of permutations: 999
##
## Model: rda(formula = sens_per_endanger %>% select(starts_with("mean_slope_")) %>% drop_na() ~ (sens_
##           Df Variance      F Pr(>F)
## RDA1       1    12.31 1.8879  0.099 .
## Residual 127   827.93
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
RsquareAdj(rda_trait_endanger_prop)
```

```
## $r.squared
## [1] 0.01464797
##
## $adj.r.squared
## [1] 0.006889293
```

Knit document

```
knit_docx(
  template_docx_file = here("docs", "PhenObs_cluster1_draft.docx"),
  knitted_docx_file = here("docs", "PhenObs_cluster1_draft_knitted.docx")
)
```

```
##
## Adjuntando el paquete: 'officer'

## The following object is masked from 'package:readxl':
##
##     read_xlsx

## [1] "Saving knitted document locally"
```

Session info

```
sessionInfo()
```

```
## R version 4.6.1 (2026-06-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=Spanish_Spain.utf8  LC_CTYPE=Spanish_Spain.utf8
## [3] LC_MONETARY=Spanish_Spain.utf8 LC_NUMERIC=C
## [5] LC_TIME=Spanish_Spain.utf8
##
## time zone: Europe/Madrid
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] pandoc_0.2.0                   officer_0.7.6
##  [3] reproduciblemanuscript_0.1.0.0 broom.mixed_0.2.9.7
##  [5] broom_1.0.13                   showtext_0.9-8
##  [7] showtextdb_3.0                 sysfonts_0.8.9
##  [9] vegan_2.7-5                    permute_0.9-10
## [11] corrplot_0.95                  ggrepel_0.9.8
## [13] progress_1.2.3                 countrycode_1.8.0
## [15] furrr_0.4.0                    future_1.70.0
## [17] gridExtra_2.3.1                readxl_1.5.0
## [19] rgbif_3.8.5                    taxize_0.10.1
## [21] ggeffects_2.3.2                glmmTMB_1.1.14
## [23] sf_1.1-1                       lubridate_1.9.5
## [25] forcats_1.0.1                  stringr_1.6.0
## [27] dplyr_1.2.1                    purrr_1.2.2
## [29] tidyr_1.3.2                    tibble_3.3.1
## [31] ggplot2_4.0.3                  tidyverse_2.0.0
## [33] here_1.0.2                     readr_2.2.0
##
## loaded via a namespace (and not attached):
##  [1] RColorBrewer_1.1-3  rstudioapi_0.19.0   jsonlite_2.0.0
##  [4] datawizard_1.3.1    magrittr_2.0.5      farver_2.1.2
##  [7] nloptr_2.2.1        rmarkdown_2.31      fs_2.1.0
## [10] ragg_1.5.2          vctrs_0.7.3         minqa_1.2.8
## [13] askpass_1.2.1       htmltools_0.5.9     haven_2.5.5
## [16] curl_7.1.0          cellranger_1.1.0    parallelly_1.48.0
## [19] KernSmooth_2.23-26  plyr_1.8.9          sandwich_3.1-1
## [22] zoo_1.8-15          TMB_1.9.23          uuid_1.2-2
## [25] whisker_0.4.1       lifecycle_1.0.5     pkgconfig_2.0.3
## [28] Matrix_1.7-5        R6_2.6.1            fastmap_1.2.0
```

```
## [31] rbibutils_2.4.1     digest_0.6.39       numDeriv_2016.8-1.1
## [34] rprojroot_2.1.1     textshaping_1.0.5   labeling_0.4.3
## [37] timechange_0.4.0    httr_1.4.8          mgcv_1.9-4
## [40] compiler_4.6.1      proxy_0.4-29        bit64_4.8.2
## [43] withr_3.0.3         S7_0.2.2            backports_1.5.1
## [46] DBI_1.3.0           MASS_7.3-65         openssl_2.4.2
## [49] rappdirs_0.3.4      classInt_0.4-11     oai_0.4.0
## [52] tools_4.6.1         units_1.0-1         otel_0.2.0
## [55] ape_5.8-1           zip_3.0.0           glue_1.8.1
## [58] nlme_3.1-169        grid_4.6.1          cluster_2.1.8.2
## [61] generics_0.1.4      gtable_0.3.6        tzdb_0.5.0
## [64] class_7.3-23        data.table_1.18.4   hms_1.1.4
## [67] utf8_1.2.6          xml2_1.6.0          pillar_1.11.1
## [70] vroom_1.7.1         splines_4.6.1       lattice_0.22-9
## [73] bit_4.6.0           tidyselect_1.2.1    knitr_1.51
## [76] reformulas_0.4.4    crul_1.6.0          xfun_0.59
## [79] stringi_1.8.7       lazyeval_0.2.3      yaml_2.3.12
## [82] boot_1.3-32         evaluate_1.0.5      codetools_0.2-20
## [85] httpcode_0.3.0      cli_3.6.6           systemfonts_1.3.2
## [88] Rdpack_2.6.6        Rcpp_1.1.1-1.1      globals_0.19.1
## [91] parallel_4.6.1      prettyunits_1.2.0   lme4_2.0-6
## [94] listenv_1.0.0       scales_1.4.0        e1071_1.7-17
## [97] insight_1.5.2       crayon_1.5.3        rlang_1.2.0
```